scientific reports

Check for updates

OPEN

Comparative analysis of deep learning models for crack detection in buildings

S. Siva Rama Krishnan[1], M. K. Nalla Karuppan[2], Adil O. Khadidos[3], Alaa O. Khadidos[4,9], Shitharth Selvarajan[5,7,8]✉, Saarthak Tandon[1] & Balamurugan Balusamy[6]

Life-time of the buildings is generally challenged by the act of nature. In-spite of the fact that the constructions provide minimum guarantee on quality and durability, certain mismatch in the composition of the materials, stress on the building, and chemical or physical imbalance of the materials, lead to surface crack. Cracks are also generated due to the shuffle of climatic conditions, which leads to the contraction and expansion of the building surfaces, and other damages. The guarantee on building safety and serviceability depends on how these buildings are successfully assessed and maintained. The development of Artificial Intelligence (AI) techniques, provide favourable solutions in-order to handle, manage and solve building cracks, through analysis using deep image neural network models, that perform classification of the building with crack images. As a result, a critical challenge for many civil engineering applications is the precise, quick, and automated identification of cracks on structural surfaces is addressed with the solutions provided by the deep image neural networks. In this research, we tackle the research gap and data scarcity by developing and curating a novel deep learning image processing for detecting cracks in brickwork. We also train and validate several deep learning models to classify brickwork images as either cracked or normal. The dataset of the proposed work contains 24,000 images which are classified through binary classes. These classes are generated for crack and non-crack images. The various parameters such as Batch size, Pooling, Activation functions Learning-rate, Kernel-Size, Normalization, and Optimizers are used for the evaluation of the model. The proposed work performs a comparative analysis of four deep image models such as Inception V3, VGG-16, RESNET-50 VGG-19, Inception ResNetV2 and CNN-RES MLP. With the analysis of all these models, the Inception V3 provides the best of all with the accuracy value of 99.98%. The InceptionV3 tops the Precision value of 99.99% and RESNET-50 tops the Recall value of 99.98%. The IncpetionV2 provided the best of the Region of Convergence value of 0.9999 which is the best among all the models for reliable and stable performance.

Keywords Building cracks, Inception V3, VGG-16, ResNet-50, InceptionV2ResNet, CNN

Different crack patterns commonly appear in buildings during construction, completion when they are subjected to extra load, or even later on in their service lives[1]. When a building component's tension surpasses its capacity, cracks begin to form. The building component may become stressed due to internal factors like heat movements, moisture changes, elastic deformation, chemical action, or external forces like wind, seismic loads, foundation settlement, etc. One of the most significant factors[2]in building cracking is thermal movement. Almost all materials expand slightly upon heating and compress slightly upon cooling. Temperature differences and consequent thermal movement are caused by changes in the ambient temperature and the variable rates at which various parts of a structure lose heat through hydration. Creep is the gradual, time-dependent deformation that occurs with sustained stresses on a concrete structure. This might create too much of tension and causes a crack

[1]School of Computer Science and Information Systems, Vellore Institute of Technology, Katpadi, Vellore 632014, Tamilnadu, India. [2]Balaji Institute Modern Management, Sri Balaji University, Pune 411033, India. [3]Department of Information Technology, Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah, Saudi Arabia. [4]Department of Information Systems, Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah, Saudi Arabia. [5]Department of Computer Science, Kebri Dehar University, Kebri Dehar, Ethiopia. [6]Student Engagement, Shiv Nadar University, Delhi-NCR 201314, Delhi, India. [7]Department of Computer Science and Engineering, Chennai Institute of Technology, Chennai, India. [8]Centre for Research Impact & Outcome, Chitkara University, Punjab, India. [9]Center of Research Excellence in Artificial Intelligence and Data Science, King Abdulaziz University, Jeddah, Saudi Arabia. ✉email: ShitharthS@kdu.edu.et

to form. There are many standard techniques[3] incorporated for prevention of cracks. One such technique is the quality and maintenance of Masonry units. After being removed from kilns, burned clay bricks and other burnt clay products shouldn't be used in construction for at least 2 weeks in the summer and 3 weeks in the winter. At that point, the crack must directly be exposed to the air.

It is best to use super-sulphated cement or to use more cement in the mortar when using bricks that contain an excessive number of soluble sulphates. Gypsum plaster (CaSO4) shouldn't be utilised for exterior or interior projects where there is a chance of moisture build-up. It is important to keep in mind that gypsum and cement cannot be combined because moisture causes a dangerous chemical reaction. The building movement can be caused by flaws, changes in ground conditions, foundation settlement and changes in the building fabric. The most typical reasons for construction cracks are[4], Clay shrinkage, land slide, vibration, subsidence, settlement, heave, and sway are some common reasons. Concrete erosion from chemical pollutants, soft clay brick degradation causes foundation failure. Building fabric deterioration brought on by woodworm, rust, wet rot, and dry rot. Expansion or contraction of materials due to moisture migration. The presence of vegetation, malfunctioning or damaged drains, flooding, damage to the building fabric, and humidity can all result in excessive water penetration. Building shear cracks can be caused by uneven bearing pressure placed on various areas of the structure or by the foundation's substantial differential settlement. When the metal ties holding the exterior skin of both cavity and solid walls collapse, there is a wall tie failure.

Deep learning, a branch of machine learning, employs multilayered neural networks known as deep neural networks[5]to mimic the intricate decision-making capabilities of the human brain. The primary distinction between deep learning and machine learning lies in the architecture of their neural networks. Experienced inspectors conduct manual inspections, which are time-consuming and depend on their empirical and subjective knowledge. This prolonged process leads to delays that further jeopardize the infrastructure's structural integrity. To overcome this limitation, an autonomous crack detection method[6] based on deep learning (DL) using convolutional neural network (CNN) techniques can be deployed.

Automated inspection for crack detection and depth estimation is crucial for assessing damage and determining the appropriate repair methods in concrete structures. Most studies[7] utilizing deep learning models for automatic inspection focus primarily on detecting and estimating the width, length, area, and direction of cracks.

Using CNNs and transfer learning models, several ensemble learning models are developed, combining self-designed CNNs, transfer learning CNNs, and hybrid models. Ensemble learning strategies[8] like weighted average, stacking, Adaboost, Gradient Boosting, and XGBoost are utilized to enhance prediction accuracy and improve generalization.

This study[9] proposes a hybrid deep learning method for pavement crack semantic segmentation, leveraging knowledge transfer between Class Activation Maps (KTCAM) and an encoder-decoder segmentation network (KTCAM-Net) using a revised CAM strategy.

A structural fault poses a significant safety hazard to the proper functioning of a structure. Therefore, cracks must be detected and repaired before they spread and lead to the structure's demolition. One well-known technique for identifying damage and faults is optimization-based methods[10]. These methods use optimization algorithms in an iterative process to minimize an objective function. These algorithms are highly efficient for damage recognition and are widely used in identification problems.

This research[11] explores the use of metaheuristic optimization algorithms to improve the predictive accuracy of the Extreme Gradient Boosting (XGBoost) machine learning model for estimating the punching shear capacity of Fiber Reinforced Polymer (FRP) reinforced concrete slabs without shear reinforcement. A database of 610 experimental results is compiled, and the effectiveness of twenty metaheuristic algorithms in optimizing the hyperparameters of the XGBoost model is evaluated. The enhanced models are then assessed and compared with existing prediction algorithms, demonstrating superior accuracy.

Motivation of the work

The most typical methods for spotting deterioration to a structure have traditionally been visual examinations. Unfortunately, these inspection methods are frequently insufficient for determining the condition of a structure, particularly when the damage is concealed from human sight. Using neural networks to forecast and assess structural degradation has drawn more attention in recent years. Artificial Neural Networks may be used to determine damage detection as an inverse issue from the measured responses to the structure's stimulation. The neural network may be taught to distinguish between the traits of a structure with undamaged parts and those of a structure with elements that have undergone varied degrees of damage. The trained neural network will then have the capacity to locate and gauge the degree of damage to specific components. AI is used to produce automatic crack detection systems that accurately identify cracks[12] and do not mistake them for minor scratches. As a result, the technicians must watch the videos frame by frame. Per-pixel window sliding is a method that may be used to find cracks in a surface. This procedure is thought to be incredibly inefficient and might raise costs significantly. The Convolutional Neural Network (CNN) may be used for pre-selection purposes to assist partition the full-size picture into a few image blocks as an alternative to employing the per-pixel window-sliding. Cracks detection is also adequately understood using inspection enhancement techniques. This work compares and analyses the different deep learning image mechanism for efficient crack detection in buildings. The contributions of the paper are,

Using the cutting edge deep neural imaging models for the assessment and analysis of the building cracks in an efficient manner. To propose a novel methodology in order to reduce the overfitting and improve the crack detection level accuracy. The proposed work employs various ImageNet models, out of which Inception V3 achieves best accuracy levels of 0.9998. The proposed model employs hyper-parameter evaluation for efficient analysis of the performance of the deep image neural models. In order to test and validate the proposed

methodology, various performance metrics like Accuracy and Region of Convergence (ROC) have been used. Comparison of the proposed methodology with existing systems, such as InceptionV2 ResNet have been analysed for testing and validation of the results.

Organization of the paper

The rest of the paper is organized as, Sect. 2 describes the Literature Review, Sect. 3 deals with materials and methods used for crack detection. Section 4 describes the illustrates System Model Architecture, Sect. 5 describes the results of various models such as inception V3, VGG-16, RESENET-50, Inception V2ResNet, VGG-19 and CNN-ResMLP with the discussion. Section 6 discusses the case-study. And, finally Sect. 7 concludes the paper with future scope.

Literature Survey

Concrete buildings exhibit cracks which are a sign of structural instability, which demands early discovery and repair as it is a critical task for ensuring safety and extending the lifespan of infrastructure. Manual inspections, which are using the traditional methods for finding cracks in concrete is a time taking process. It also has limitations in-terms of accuracy. Nowadays with the advancement of deep learning methodologies, it is possible to use computer vision algorithms to detect cracks in concrete.

Machine learning techniques for Crack detection

Parhi and Choudhury[13] reviewed various smart crack detection methodologies used in different types of structures, including vibration-based, strain-based, and electrical resistance-based methods. Authors explained about how support vector machines and neural networks are used as machine learning algorithms for crack detection and emphasized the significance for additional study to provide more precise and effective crack detecting methods.

Silva and Lucena[14] researched on a machine learning model with a deep learning-based image classification approach for concrete crack detection. Authors used a dataset of 900 images to train the model and achieved an accuracy rate of 92.27% and concluded that deep learning-based approaches are effective in detecting cracks in concrete structures.

Hoang[15] proposed an image processing technique with an improved Otsu method and M2GLD method for image thresholding to detect surface cracks in building structures, which provides a better accuracy for crack detection.

Kapadia et al.[16] discussed the computer vision technology in order to monitor cracks in concrete surfaces or buildings. The proposed methodology used edge detection and blob analysis techniques to detect cracks in concrete surfaces and the model achieved the accuracy of 94% with Inception V3 model.

Mohan and Poobal[17] examined various image processing techniques like Electroluminescence Image, UAV Camera Image, SEM image, Radar Image, Probe Image, Microwave Image, Sensor Image and Video Image. A total of 41 research papers were analysed for the comparative analysis.

Golding et al.[6] used Convolutional Neural Network (CNN) model for crack detection, where 40,000 RGB images were processed before training a pre-trained VGG16 architecture to create various CNN models. Authors also used the grey-scale models which resulted an F1 score of 99.331% for 10 epochs: 99.549% for 20 epochs.

Nishikawa et al.[18] proposed two methodologies for better and accurate concrete crack detection. First, with the help of genetic programming (GP) to create an image filter for the large crack detection and the second, the iterative application of the image filter around the cracks for removal of residual noise after detection of small cracks, and with the help of both of methodologies the image processor take only 10 s to process 1.5 million pixels.

Dung and Le Duc[19]. evaluated fully Convolutional Network (FCN) for a semantic segmentation on concrete crack images. Authors orated three alternative pre-trained networks, which are evaluated for image classification on a public concrete crack dataset which consist of 40,000 227×227 pixel images. Out of which 500 annotated 227×227 pixel crack labelled images were used to train the encoder and decoder for FCN network. In this work, an accuracy of 90% was achieved.

Cheon et al.[20] devised a method for detecting surface cracks in concrete structures using image processing and emphasized on transforming the RGB colour image to a grayscale image and enhancing the cracks with a Laplacian of Gaussian (LoG) filter. The Hough transform was used to further analyse the discovered cracks and produce precise crack locations and dimensions. The results of the experiments demonstrated that the suggested method was successful in identifying surface fractures in concrete structures.

Cho et al.[21] presented using terrestrial laser scanning in conjunction with the octree structure an image processing method for fracture identification in concrete constructions and also identified crack locations by splitting the point cloud data into octree voxels and used a density-based clustering algorithm and concluded that large cracks as well as the smaller cracks were detected even if the image was compressed up to 50%.

Yamaguchi and Hashimoto[22] suggested a quick crack identification technique employing percolation-based image processing for large-size concrete surface pictures. In the suggested method, the RGB colour image was converted to a binary image, and the cracks are found using a percolation-based algorithm. The experimental findings demonstrate that the suggested method was successful in identifying cracks in photographs of large-scale concrete surfaces.

Wang and Huang[23] carried out a comparison investigation of the newest methods for visualising cracks in concrete structures. In the study, edge detection, threshold, and morphological techniques were investigated for their applicability in detecting fractures in concrete buildings.

Zhao and Liu[24] researched on a novel crack feature pyramid network which was based on image processing. The proposed method reduced computational cost. Authors used publicly available pixel-level picture datasets to train the network.

Yamane[25] utilized a post-processing step to remove false positives after using a U-Net architecture to segregate crack areas from input photos. Authors tested images in various conditions including shadow, dirt and it was discovered that not only the fracture region could be recognized but also the trace of tie-rod holes and formworks could be removed from the detection result with high accuracy by using images of concrete structures with a variety of challenging situations.

Arbaoui[26] developed a CNN to each band of the input photos after using a wavelet transform to break them up into distinct frequency bands in order to find cracks. The methodology achieved higher accuracy by combining major data analysis tools i.e. wavelets and deep learning and achieved accuracy of 90%.

Deep learning techniques for crack detection

Ko et al.[27] proposed an automated building exterior crack inspection system with the help of UAVs, operated with the support of the opensource deep learning, and photogrammetry. The system can accurately detect and classify cracks in building exteriors with 100% success percentage in building, 80% in walls, providing a cost-effective and efficient alternative to traditional inspection methods.

A crack detection technique incorporating convolutional neural networks was proposed by Qayyum et al.[28] with variable image datasets. The model was not able to achieve a higher accuracy and authors tested 3 dataset models which produced the overall accuracy percentages of 48.30%, 64.30% and 65.30% respectively.

Kolappan Geetha et al.[29] proposed a fast detection method for missing thin propagating cracks using deep learning based concrete crack/non-crack classification and successfully combined the benefits of picture binarization, a shallow 1D DL model's computational efficiency, and inherent frequency-domain feature extraction characteristics of the cracks in the Fourier-based DL model (1D-DFT-CNN). The research concluded that, it takes less time and effort to restore pixelated labelling as taken in the current semantic segmentation.

Munawar et al.[30] compared and reviewed various image-based crack detection methods, including threshold-based, edge-based, and region-based methods. Authors highlighted the advantages and limitations of each method, and suggests that deep learning-based methods render higher accuracy in crack detection and also discussed the importance of selecting appropriate datasets and training parameters to achieve higher accuracy rates. The results recorded vary between 75 and 100%.

Ali et al.[31] assessed the performance of deep CNN-based crack for concrete structures and compared the performance of 5 CNN models including VGG16, VGG19, InceptionV3, ResNet-50 and customized CNN for crack detection and concluded that the customized CNN model, VGG-16, InceptionV3 model gave better results with the small dataset whereas, VGG-19 give optimized results for larger datasets. Authors concluded that training a customized CNN model is the best option, since with small amount of data it provides high performance.

Li et al.[32] suggested a deep convolutional neural network based crack detection technique. Authors also researched about modified CNN with AlexNet through 6000 images were trained and evaluated validation accuracy for various learning rates and determined 0.01 as the best learning rate with maximum validation accuracy of 99.06%. Authors also trained the CNN into a smartphone application for enabling the general people to spot cracks.

Kim[33] proposed a cutting-edge method for detecting cracks in concrete constructions utilising image processing and deep learning techniques and suggested methods, that entail using a sizable collection of photographs of cracked and uncracked concrete structures to train a convolutional neural network (CNN). Then, with fresh photos of concrete structures, the trained CNN was used to find fissures. The experimental results demonstrate that the proposed strategy performed more accurately and effectively as compared to the conventional fracture detecting techniques.

Comparing the proposed method to Talab's method, K-means clustering could reduce the false detection of fractures on structural joints and sediments. Adam and Ammayappan[34] developed a hybrid deep learning methodology for precisely identifying cracks in concrete constructions. This model finds and separates cracks in concrete structures. The suggested method combined a fully convolutional neural network (FCN) and a region-based FCM, which results in 10% higher accuracy as compared to other existing models.

Nguyen Kim et al.[35] suggested a technique using deep learning and image processing to automatically identify cracks. The RGB colour image was converted into a grey-scale version in the proposed method, and a convolutional neural network was then employed to search for the cracks. The experiment findings used different picture data sizes for the suggested training model. Using 3000 images for each class, the test data accuracy was recorded as 91%, and the training maximum accuracy (ACC) was 96.1%. Also, the 4000 image dataset's minimum testing accuracy was 87.5%.

Cha[36] developed deep learning crack detection model using convolutional neural network and authors used a pre-trained convolutional neural network to extract necessary features from input images and thereafter applied binary classification model to identify cracks in the concrete. Authors took over 332 photos, of which 277 had a resolution of 4,928 by 3,264 pixels for training and validation and 55 had a resolution of 5,888 by 3,584 pixels for testing and recorded 98.22% accuracy and 97.95% respectively.

Hacıefendio˘glu[37] proposed a Faster R-CNN algorithm based on deep learning for detecting concrete road cracks. The suggested method successfully detected and categorized cracks in concrete road photos with high accuracy. 323 photos with a resolution of 4128 × 2322 pixels and a 16:9 aspect ratio was used to demonstrate the efficacy of the suggested method and to confirm how well damage identification performed on the chosen concrete road surface. Author conducted an experiment that created 1128 object bounding boxes for the object detection procedure and also tested effects of the weather and lighting intensity. Photographs taken in sunny,

S. No.	Author Details	Methods used	Advantages	Limitations
1	Kapadia et al.[16]	The edge detection and blob analysis techniques to detect cracks in concrete surfaces.	Achieved accuracy of 94% with Inception V3 model	The system has high computational cost.
2	Nishikawa et al.[19]	Evaluated fully Convolutional Network (FCN) for a sematic segmentation on concrete crack images	Improved accuracy	Over fitting problem need to be addressed.
3	Kim[33]	Cutting-edge method for detecting cracks in concrete constructions	The proposed strategy performs more accurately and effectively as compared to conventional fracture detecting techniques	Lacks in explaining the outputs and discussions.
4	Cho et al.[21]	Terrestrial laser scanning in conjunction with the octree structure was used for image processing	The large cracks as well as the smaller cracks were detected even if the image was com- pressed up to 50%	Method lack in interoperability.
5	Zhao and Liu[24]	A novel crack feature pyramid network was proposed for image processing	The computational cost was reduced.	The detection accuracy needs to be enhanced.
6	He, Y.et.al[60]	Local–Global Feature Adaptive Fusion Network for Building Crack Detection	Local feature extraction, multi-feature adaptive fusion (MFAF) module is proposed to integrate local and global features from different branches and facilitate representative feature learning	Lack of hybrid models of CNN and evaluation of the performance metrics. No comparison across the models
7	Cao, J.et.al[61].	Automatic detection and assessment of crack development in ultra-high-performance concrete in the spatial and Fourier domains	The work describes that the Fourier domain exhibit distinct concentration around a central position the power spectral density of cracks exhibits an increasing trend over time.	It does not investigate the image features related to the crack and depth related to imaging
8	QiQi Zeng, et.al[62].	An Attention Crack Detection Network for Structural Health Monitoring	ACDN, deep learning-based crack detection approaches shows remarkable recognition accuracy. ACDN also exhibits superior cracks recognition capability under complex backgrounds, where there are no misrecognitions occurred.	Surface and detection accuracy is discussed. There is no hyper parameter evaluation and discussion of possible models.

Table 1. Table of essential literature review.

Ref no	Methods	Similarities	Differences
14	Deep learning model, a transfer-learning methodology was applied	Image dataset with and without cracks taken (referred paper 3500 images)	Image dataset size is huge leading to better accuracy.
32	Customized CNN and VGG-16 models	Compared the performance between VGG-16, VGG-19, ResNet-50, and Inception V3 models	The computational time and complexity of the proposed CNN model was less than the VGG-16 model
27	The combination of two major data analysis tools which are wavelets and deep learning	SDNET2018 was used for the automatic detection of external cracks.	Our proposed method uses deep image processing method for detection of internal and external cracks

Table 2. Comparative table between this proposal and studies in the literature.

overcast, and foggy weather, at sunset and during the full moon, were examined for cracks using crack detection software.

Yu[38] proposed a deep convolutional neural network (CNN) fracture detection technique for concrete structures that was augmented by a chicken swarm approach. To increase the accuracy of crack identification, the author used first extract features from the input photos using a pre-trained CNN and then utilized the suggested optimization approach. An improved chicken swarm algorithm and deep convolutional neural network was used to create the vision-based crack diagnostic tool (ECSA).

Deep learning techniques are highly applicable for detecting cracks in buildings. However, one significant challenge in this domain is the hyperparameter tuning process for convolutional neural networks (CNN). This paper aims[39] to address this issue by proposing an automated hyperparameter tuning approach for crack image classification. A public dataset containing 40,000 images of walls and floors from various buildings was utilized, with images categorized into two classes: negative (non-crack) and positive (crack). Statistical methods, including analysis of variance, the Scott–Knott method, and the HyperTuningSK algorithm, were employed for hyperparameter tuning.

Deep learning techniques have demonstrated significant potential in computer vision applications within civil engineering. A key research area is the detection and classification of construction machines using convolutional neural networks (CNNs). However, a major challenge in implementing machine learning models for vision systems is hyperparameter optimization. This paper aims[40] to propose a detailed statistical approach to hyperparameter tuning for deep learning in the visual classification of construction machines. The method for selecting hyperparameters employs statistical concepts, including analysis of variance and the Tukey test. Additionally, three research questions were formulated to explore statistical differences in performance when varying the CNN architecture, optimizer, and their configurations.

Table 1 shows the comparative analysis of literature review conducted. Table 2 depicts the similarities and differences with existing works and proposed method.

Research gaps

Considering the above literature, the following research gaps are observed in existing studies.

- The crack detection process iteration was not tuned by callbacks with the learning rate suited for the specific models.

- In some of the existing works, the optimizers were chosen randomly without the significance of impact reflected on the performance of the models.
- A strong lack of investigation of hybrid models for the crack detection process that can improve the stability of the system, and enhance the accuracy of validation.
- Some metric evaluations which are necessary for the complete understanding of the model performance are missing.
- And, there was a Lack of Study on multi-layer perception for the classification of crack images.

Materials and methods

Materials

This section deals with various types of the cracks, the underlying causes and various treatment mechanisms deployed to recover and repair the cracks, with effective implementation of various strategies mentioned below.

Types of the cracks

The following are the various types of geological crack shapes and types[41]. Figure 1. depicts the various types of cracks :

- **Hairline Cracks**: Hairline cracks are mainly due to the shrinking of the plaster. These cracks are sensitive and occur due to the presence of the timber frame. The timber frame moves with respect to the time, but the plaster don't, which leads to mild and less significant hairline crack.
- **Steeped cracks**: Steeped cracks are created over the horizontal and vertical joints in buildings, such as mortar between bricks and blocks which indicates structural movement of the building.
- **Vertical cracks**: Vertical cracks are due to the significant stress on the structure of the building due to failure of the bricks and blocks.
- **Horizontal cracks**: Horizontal cracks indicate a failure on the wall which is a serious concern regarding the safety of the wall.

Steps for crack detection

- Crack detection with ultrasonic testing: A typical ultrasonic crack detection inspection system is made up of many transducers that function as emitters and receivers in an impulse-echo mode. The incident ultrasound signals must be refracted in such a way that the sensor carrier design permits them to travel within the pipe at an angle less than 45°. Thus, slanted probes are employed. High frequency ultrasonic energy produced by the ultrasonic probe is delivered into the materials and travels through them as shear waves. A portion of the energy will be reflected by the defect surface when one exists.
- Composites reinforced with fibrous elements for monitoring structural health: Fiber-optic sensors can be used for fracture detection, localization, and crack width measurement, depending on the measuring apparatus used. Crack detection and crack width measuring are based on the physical fact that uncoupled light absorbs more when the fiber cross-section becomes weaker. The presence or altered breadth of a fracture may then be determined by measuring the attenuated light at the end of an optical cable.

Remedial measures

Before remedial procedures, the examination is required to determine whether the cracks are produced due to external factors (subsidence) or the internal fabric of the buildings. The cracks must also be examined for the static or dynamically varying so that the treatments can be planned.

The treatment for the building changes from small fabrication to major interventions. The cracks due to thermal expansions and vibrations ever exist. The remedial methods must be chosen carefully such as the one with the ability to deform as plastic manner rather than the one which may develop further strain around the cracks. Hydraulic and fat-limes provides enough plasticity between the crack surfaces. Non corrosive steel sheets can be also used as the crack prevention solution, which would be able to withstand some amount of excess strain that is applied on the building. It redistributes the stress to wide area and prevents further damage.

Instruments such as boroscopes are used for the examinations on the movement of the crack, Grouting the gaps of the cracks with the cementitious material that can also deal with the underlying cause.

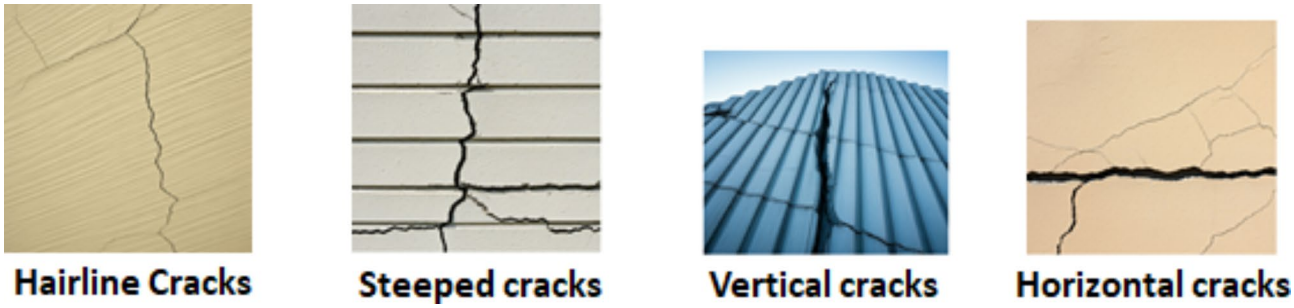


Fig. 1. Types of the Cracks.

Dataset

Surface crack Detection Dataset[42] used in the proposed work contains images of various concrete surfaces with and without crack. The image data is classified as positive and negative with binary classification for the presence of crack or no-crack respectively. Every class has 20,000 images with a total of 40,000 images with 227×227 pixels with RGB channels. The dataset is generated from 458 high-resolution images (4032×3024 pixels). The contrast in the high quality images had huge variation with respect to surface quality and the illumination. No data augmentation in terms of random rotation or flipping or tilting is applied. This dataset is used for the binary classification with 0 representing no-crack class and 1 representing the crack class images. Since the number of images are equally distributed in the dataset, it does not suffer from the class imbalance issues. The predictions on the dataset are unbiased. The images are pre-processed with sampling with DCT application. The images are sampled with feature size of 224*224*3 for some models and 299*299*3 for some ImageNet models. The feature selection is done by Max and Average Pooling, The training batch processing is done based on the number of samples and the batch size.

Methods

This section covers up various models used in the proposed work, with emphasis on the construction, design and layered architecture. This section also deals with technical description of each model in detail.

Inception V3

Inception v3[43] is a deep convolutional neural network architecture, that was developed by the google researchers in 2014 and won the ImageNet Large Scale Visual Challenge (ILSVC) competition. It provides better efficiency and accuracy than the predecessors. This model contains auxiliary classifiers, factorized convolutions for cost reduction and better performance, and efficient grid size reduction for reduction of the size of the feature maps. This model is designed for image classification and it helps to reduce the number of parameters in the network and improve the process effectively. There are 5 types of layers.

- Input Layer which takes an image with the dimensions (224, 224, 3) and the number of colour channels are Red, Green, Blue. Pooling layers and stride convolutions are used to compress the spatial dimensions of the image, which reduces the computational cost of the network.
- Stem Layers performs convolutional and pooling operations to reduce spatial dimension and increases the maps which are featured.
- Inception module has several parallel branches, including convolutions, pooling, and 1×1 convolutions, that carries out various actions from the input.
- Auxiliary classifier purpose is to provide additional supervision during training and to enhance the performance of the network.
- Output Layer is a fully connected layer that produces the probabilities for the input image by the result of previous layers.

Strengths and weaknesses.

Several methods were employed by the Inception V3 model to optimize the network in order to improve model adaption.

- • It operates more efficiently.
- • Its speed isn't affected by having a deeper network than the Inception V1 and V2 models.

One drawback is that because of its intricate design, training takes a lot of time and computational resources. Another drawback is that, while pretrained ImageNet weights are frequently required for transfer learning, it could not function well on smaller datasets.

VGG-16

VGG is known as Visual Geometry Group (VGG) and is a deep convolutional neural network architecture utilized for image classification problems. VGG 16[44] model is a type of CNN model was introduced by K.Simonyan and A.Zisserman through the research paper titled "Very Deep Convolutional Networks for Large- Scale Image recognition" published by the oxford press. This model achieved superior accuracy and served as a benchmark for the other image processing models also. The term 16 refers that the model contains 16 layers.

It has 16 layers, including 3 fully linked layers and 13 convolutional layers. VGG16 is efficient at learning intricate patterns in images because of its deep design and numerous parameters. Convolutional layers make up the first 13 layers. Edges and corners are detected by the first few layers, forms and textures are detected by the following layers, which have more complicated characteristics. Each one has a 3×3 kernel, a stride of 1, and a rectified linear unit (ReLU) activation function. The size of the input image is decreased by the fact that the top two layers also have a stride of 2. Spatial dimensions of the feature maps are down-converted in the following five max pooling layers. The last layer, a SoftMax layer that generates the classification output, is one of the final three completely connected layers. Image input layer: $224 \times 224 \times 3$ Convolutional layer with 64 filters, each with a 3×3 kernel and ReLU activation Max pooling layer with a 2×2 pool size and stride of 2, as well as a convolutional layer with 128 filters, each with a 3×3 kernel and ReLU activation Max pooling layer with a 2×2 pool size and a stride of two Convolutional layer with 256 filters, each with a 3×3 kernel and ReLU activation Convolutional layer with 256 filters, each with a 3×3 kernel and ReLU activation.

Strengths and weaknesses: Accurate categorization is made possible by VGG16's capacity to extract high-level characteristics from pictures. The computational complexity and abundance of parameters in VGG16, however, can make training and inference time-consuming. This is a drawback of the method.

ResNet50

ResNet-50 is a deep neural network architecture, called as Residual Network[45], which is 50 layers deep. This model is a popular choice for computer vision problems due to the depth and the efficiency. The "Res" part indicates that the architecture uses the residual blocks for building the network. These blocks reduce the vanishing gradient descent problems, due to the deep training and convolution. In order to combat the degradation issue in deep neural networks, wherein the addition of more layers can result in decreased performance, it makes use of residual connections. For several computer vision applications, including object detection, semantic segmentation and picture classification, ResNet-50 has been extensively employed. Resnet 50 layers: -.

- Max pooling and a convolutional layer for input processing.
- Convolutional layers in four stages, each containing a number of residual blocks.
- Multiple convolutional layers with skip connections that omit one or more layers make up the residual blocks, which keep the input signal intact.
- A layer of global average pooling that combines the final residual block's output.
- A layer with complete connectivity that generates the categorization output.

Strengths and weaknesses: It can perform image categorization jobs with high degrees of accuracy. The possible disappearance of gradients in extremely deep networks, which might cause the gradient descent process to lag, is a drawback of utilizing ResNet. This may make it more difficult for the network to learn, particularly as the number of layers rises.

Convolutional neural network (CNN)

Convolutional Neural Networks (CNNs)[46] is a category of a deep neural network that is most commonly applied for the problems related to the computer vision and the imagery. CNN is extremely successful in the application areas of the image recognition, image classification, object detection, with various relevant application regarding the computer vision. CNNs are designed to learn with automation and adaptability with spatial hierarchies of features from input images.

For image classification, object detection, and other computer vision applications, a CNN is a type of deep neural network that is frequently utilised. It's effective because it uses the spatial correlations between adjacent pixels in the image and learns spatial properties from raw pixel data. Additionally, the number of learnable parameters in the network is significantly decreased by employing shared weights throughout the entire image, enabling the training of very deep networks with numerous layers. The model contains eight input layers, 16 polling layer, 32 Dense layer and one output layer with Sigmoid activation function to determine the output.

The various operations of CNN are described as,

• CNN is a type of deep neural network primarily used for image-related tasks, such as image classification, object detection, and image segmentation.

- CNNs are designed to automatically and adaptively learn spatial hierarchies of features from input data.
- Key components of a CNN include convolutional layers, pooling layers, fully connected layers, and activation functions like ReLU.
- Convolutional layers use filters to detect features like edges, textures, and patterns in the incoming data.
- The feature maps are down-sampled by pooling layers, which lowers their spatial resolution and computational difficulty.
- The weights of the filters are learned by CNNs using back-propagation and optimisation methods like gradient descent, which also minimise a loss function.

Strengths and weaknesses: One of the best methods for precise and efficient image processing is CNN. However, in order to attain high accuracy rates, it needs a lot of compute power and big datasets. Figure 2is representing the CNN architecture,the evolution of the CNN models are discussed in the paper[47].

VGG19

VGG 19 model is[48] a deep convolutional neural network model, which can classify millions of images, with around 1000 classes. It has totally 19 layers, in which 16 layers are convolutional layers and three layers are fully connected layers. It acquires image data with the dimensions of 224*224*7. This is used for the image classification problem. This is similar to VGG-16 with enhanced layers for convolution and transfer learning. The architecture of VGG is illustrated in Fig. 3. The input kernel size presented in the model is 224*224*3 with 8 inputs. The input layer followed by the polling layer which uses the MAX Pooling technique. The size of the kernel here is reduced to 109*109*8. We apply the 2D convolution layer, then the crack images are applied with the transfer function. At last, the output layer is optimized with RELU function that classifies a crack or non-crack image as 1 and 0 respectively.

There are Commonalities and dissimilarities between VGG-16 and VGG-19, enlisted as follows:

- VGG16 and VGG19 are specific CNN architectures developed by the Visual Geometry Group (VGG) at the University of Oxford.
- VGG16 consists of 16 weight layers, including 13 convolutional layers and 3 fully connected layers.
- VGG19 extends VGG16 by adding 3 more convolutional layers, resulting in 19 weight layers.

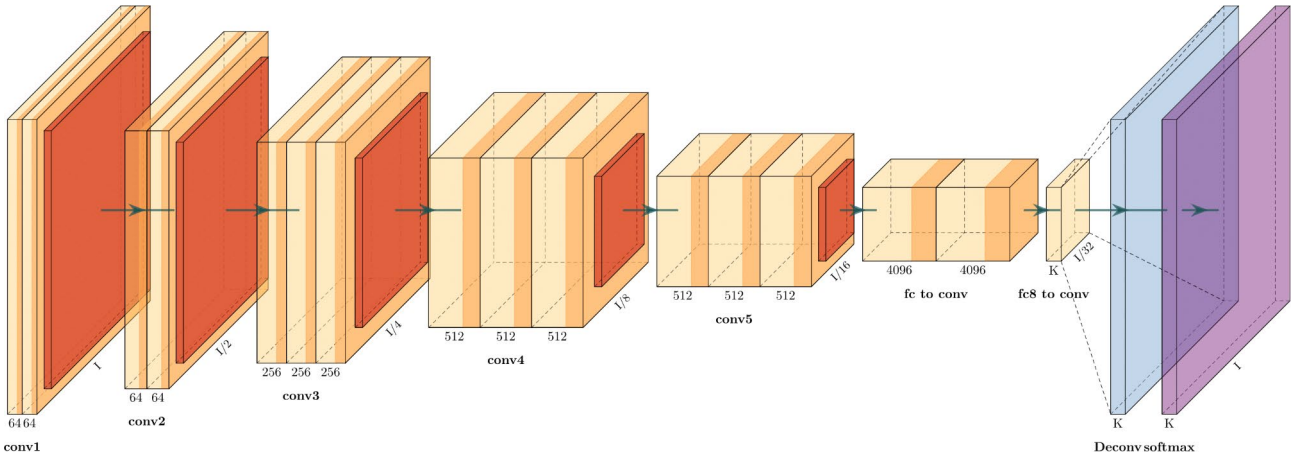


Fig. 2. A sample Neural model building for CNN.

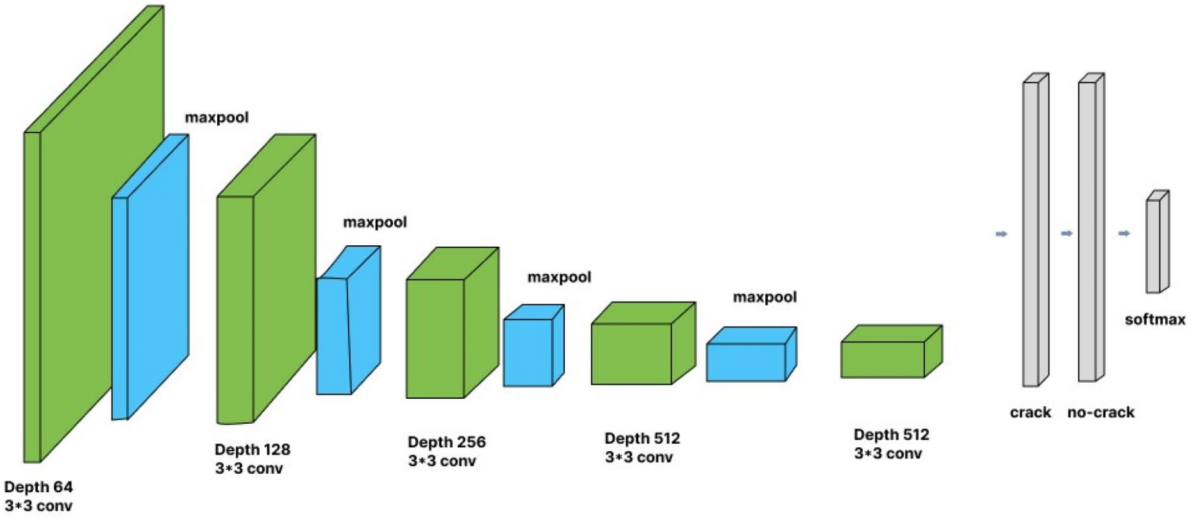


Fig. 3. A sample Neural model building for VGG-19.

- Both VGG16 and VGG19 use small 3 × 3 convolutional filters with a fixed stride of 1 and same padding throughout the network.
- They use max-pooling layers to downsample feature maps.
- VGG networks are known for their simplicity and uniform architecture, making them easy to understand and implement.
- They have been widely used as a baseline for various computer vision tasks and as feature extractors.

Strengths and weaknesses: VGG19 can achieve a higher accuracy but takes more time to train its parameters.

InceptionV2ResNet

The Inception2ResNet is[49] used for the classification in the proposed work and is capable of processing 1000 classes with the size of 210 MB. This model uses soft-max layer for activation and three ResNet layers for transfer learning process. This model is backward compatible. It uses the residual connections for learning between InceptionResNet and it is 164 layers deep. This model has 5*Inception-ResNet layer connected with the input, followed by 10*Inception-ResNet in the hidden layer and a 5*InceptionResNet layer connected to the soft-max layer for the activation.

The working principle of the InceptionV2ResNet layer is described as,

- Average pooling and a convolutional layer for input processing.
- Convolutional layers in three stages, each containing an Inception- ResNet Block.

- ResNet-Inception Layer A, B and C provides deeper convolution stages with memory and processing the batches of various images with different block sizes.
- SoftMax layer provides the activation of the target function, with specific number of classes represented in the list of the parameters.
- A layer with complete connectivity that generates the categorization output.

The salient features of the InceptionV2ResNet are:

- InceptionV2ResNet combines the Inception module with residual connections to facilitate training of very deep networks.
- It features the Inception module, which employs several filter sizes concurrently to collect various features.
- The vanishing gradient issue is solved via residual connections, which also make it possible to train very deep networks.
- It has been used for a variety of computer vision applications, including picture categorization, object identification, and facial recognition. InceptionV2 is renowned for its effectiveness.

Strengths and weaknesses: For better performance, the inception and residual connection models are combined in the InceptionResNetV2 architecture. By using a hybrid technique, the network may take use of both models' advantages, such as quicker training periods and the ability to avoid the vanishing gradient issue. One drawback, though, is that gradients may vanish in very deep networks, which might cause the gradient descent process to lag. This may make it more difficult for the network to learn, particularly as the number of layers rises.

System model architecture

The models are trained and validated with 7 epochs and batch size of 438 samples. The inceptionV2ResNet took 20 epochs with 1000 batch size for convergence[50]. The various optimizers were used by these models, such as Adam for Inception V3, InceptionV2ResNet and CNN-ResMLP. RMSProp is used for VGG-16 and VGG19, SGD for ResNet-50. These optimizers control the call back using learning rate schedulers[51] that control the batch processing and model training and validation.

The steps mentioned below, are involved in the process of encoding and decoding where an image is analysed and cracks are detected.

- An image with the dimensions (224, 224, 3), having three colour channels: red, green, and blue. Pooling layers and stride convolutions are used to compress the image's spatial dimensions, which reduces the computational cost of the network.
- Stem layers perform convolutional and pooling operations to increase the number of featured maps and decrease the spatial dimension. Only a few of the parallel branches that the Inception module provides carries out various tasks with the input are convolutions, pooling, and 1×1 convolutions.
- The auxiliary classifier's objective is to enhance network efficiency and provide extra supervision during training. After this, the result is generated in output layer (fully connected layer). The overall process architecture is represented in Fig. 4 and crack observation architecture is represented in Fig. 5. This diagram is cited from[52].

The four models that are used in this proposed work are represented in the following workflow.

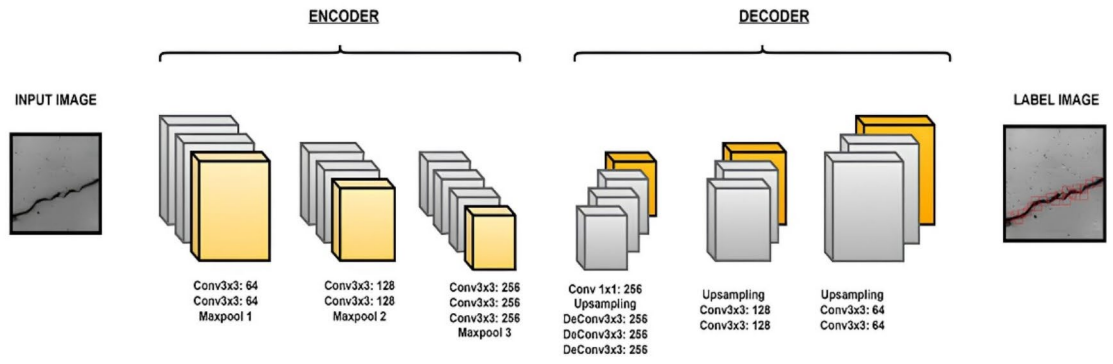


Fig. 4. Convolutional Deep Image Architecture for Crack Detection.

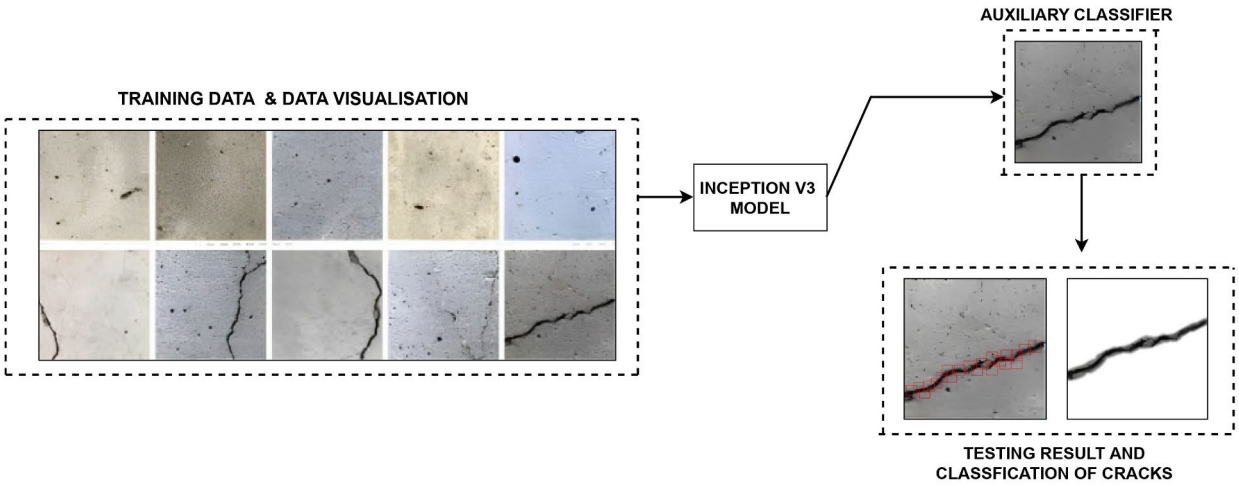


Fig. 5. Convolutional Deep Image Architecture for Crack Detection.

Convolutional layer

This layer applies a series of filters to the input image to extract features like edges, corners, and other visual patterns. A group of feature maps that indicate the activation of the filters at various points in the input image make up the convolution layer's output.

Pooling layer

The purpose of a pooling layer is to downscale the feature maps while keeping the most crucial features. This can aid in preventing over-fitting and lowering the network's computing cost. The typical method for how a pooling layer operates is to divide the input feature map into non-overlapping rectangular sections. Pooling layers come in two different varieties as per Eq. 1 as max pooling and Eq. 2 as average pooling. Max pooling: - This decreases the spatial size of the data and aids in information retention.

$$Output[i][j] = max(input[k][L]) \tag{1}$$

and produced the output matrix by moving over the input matrix, taking the maximum value in each region as,

$$Output[i][j] = avg(input[k][L]) \tag{2}$$

Produced the output matrix by moving over the input matrix, taking the average value in each region.

Activation layer

The network can learn non-linear correlations between the input data and the output by applying a non-linear function to the output of the preceding layer in the activation layer. Activation layer has 3 types they are: -.

Rectified Linear Unit (ReLu)

The ReLU function's output is only the highest value between 0 and its input, x. The output is identical to the input if the input is larger than 0. The output is 0 if the input is less than or equal to 0 represented as per Eq. 3.

$$f(x) = max(0, x) \tag{3}$$

where *x* is the input value and 0 is the maximum output value.

Sigmoid

It is defined and mathematically represented as per the Eq. 4.

$$f(x) = 1/(1 + e^{-x}) \tag{4}$$

where *f(x)* is the result and x are the output of the function and The advantage of the sigmoid function is that its output is always in the range of 0 and 1, making it suitable for modelling probability and binary classification issues.

Hyperbolic tangent

The tangent function is represented as per Eq. 5.

$$tanh(x) = (e^{x} - e^{-x})/(e^{x} - e^{-x}) \tag{5}$$

where x is the function's input and $tanh(x)$ is its result. Any input value is converted by the *tanh* function to a value between – 1 and 1.

Fully connected layer

When a layer is fully linked, its input is a vector of length and that represents the output of the layer that came before it, and its output is another vector of length m, where m is the number of neurons in the layer. The full input vector is received by each neuron in the layer, which then provides a scalar output that is merged with the outputs of the other neurons to create the final output vector which is represented as per the Eq. 6.

$$Y = WX + B \tag{6}$$

where Y is the output of the fully-connected layer and X is the input, W is the weight function and B is the bias. All the methods that use batch-normalization which is represented as per the Eq. 7.

$$Y_{b,c,d,e} = X_{b,c,d,e} + \mu_{c} / \left(\sqrt{\sigma_{c} + c}\right) + \beta_{c} \tag{7}$$

Where Y and X are the batch functions of the output and input respectively. b, c,d, e are batch, channel, input and output functions as per order.μ represents mean and σ represents the standard deviation. The hyper-parameter evaluation of all the models is discussed in the Table 3.

Optimization algorithm

All the deep learning models use an algorithm to map the input and the output with an optimization algorithm. These algorithms enhance the performance of the model and eliminates the error rates even if it is working on the unseen data. These algorithms reduce error rates when mapping to the output. The proposed work used Adam, SGD and RMS prop architectures for supporting the base-line deep neural network models.

Dropout

Dropout layer is added to avoid overfitting of the classification. This layer is generally avoided by the training process like a non-existing entity. Dropout layer enhances the reliability of the classification by preventing the overfitting of the classification process.

Kernel size

The kernel size determines the speed of transition in the model progression This can optimally determine the best fit the data through the kernel density calculation and connection.

The overall process is illustrated in Flowchart shown in Fig. 6 and overall process algorithm is presented in Algorithm 1.

Method	Pooling	Normalization	Activation	Dropout	Optimizer	Kernel Size
Inception V3	Max	Batch	RELU	Yes	Adam	150*150*3 kernels
ResNet-50	Average	Batch	RELU	Yes	SGD	150*150*3 Kernels
VGG-16	Average	Batch	RELU	Yes	RMS prop	224*224*3 Kernels
CNN-ResMLP	Average	Batch	GELU/SIGMOID	No	RMS prop	300*300*3 kernels
InceptionV2	Average	Batch	SoftMax	Yes	Adam	224*224*3 kernels
VGG-19	Average	Batch	Sigmoid	Yes	Adam	224*224*3 kernels

Table 3. Hyper parameter analysis.

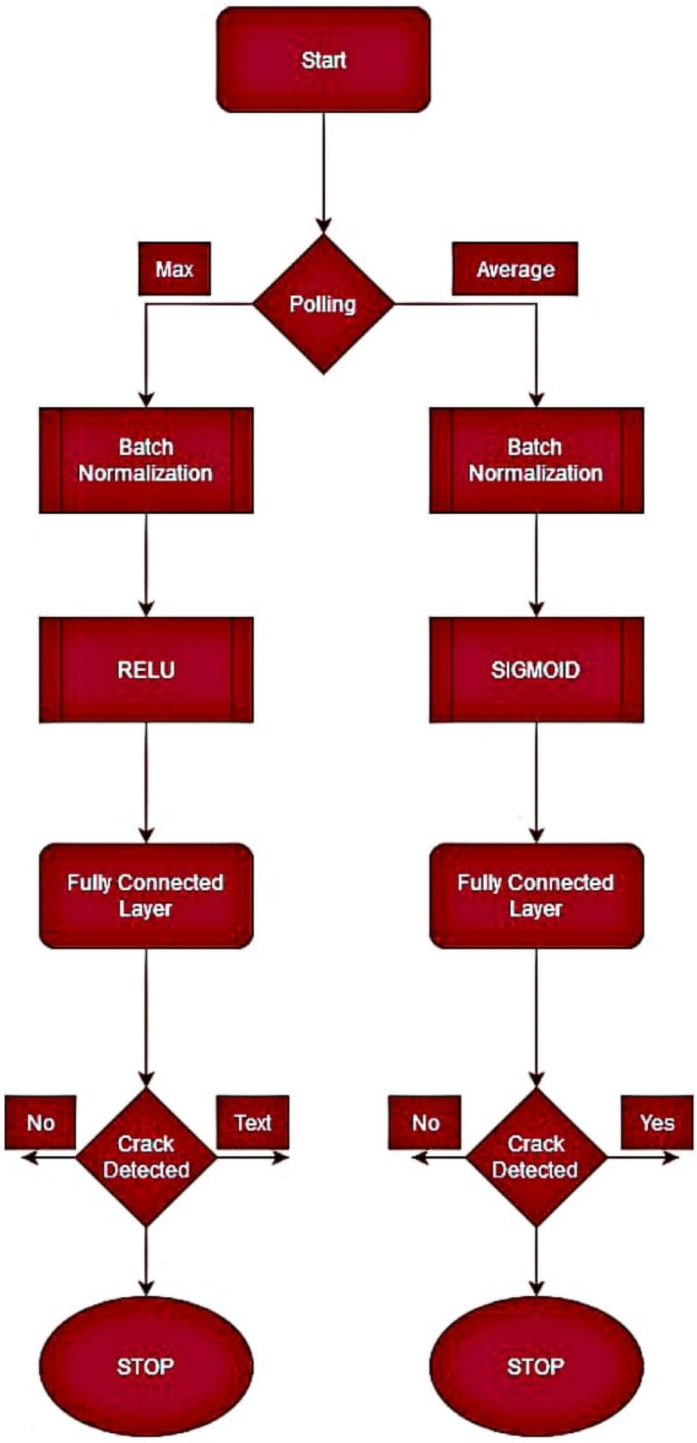


Fig. 6. Overall process flowchart of the convolution process.

Data: $x = [\sum_{n=0}^{m} X_n]$;
Result: $y = wx + b$;
$y \leftarrow ytrain,ytest$;
$x \leftarrow xtrain,xtest$;
$n \leftarrow imagesamples$;
$T_P \leftarrow TruePositive$;
$T_N \leftarrow TrueNegative$;
$F_P \leftarrow FalsePositive$;
$F_N \leftarrow FalseNegative$;
$w \leftarrow Weights$;
$b \leftarrow Bias$;
$J \leftarrow Loss\ co\text{-}efficient$;
$Features\ of\ Evaluation \leftarrow i,j$;
Max-Pooling:
$max[i][j] = max(input[k][L])$;
(or)
Average-Pooling:
$avg[i][j] = avg(input[k][L])$;
Normalization:
$xtrain = xtrain - E[xtrain]/\mathrm{p}(xtrain)$;
Activation Relu: $f(x) = max(0,x)$;
(or)
Activation function Sigmoid:
$tanh(x) = (e^x - e^{-x})/(e^x + e^{-x})$;
acc ← Accuracy;
while $xtest\ != 0$ **do**
if x_{test} *Is* ***crack*** **then**
$acc \leftarrow T_P + T_N /(T_P + T_N + F_P + F_N)$;
$loss \leftarrow J(w^T,b) * 1/m$ P $_{m\,1}(y^i\ y^j)$;
Classification of Crack Images
x_{test} is **no-crack** $acc \leftarrow T_P + T_N /(T_P + T_N + F_P + F_N)$;
$loss \leftarrow J(w^T,b) * 1/m$ P $_1^m(\bar{y^i},y^j)$;
end

Algorithm 1. Algorithm for the overall image classification problemExperimentation, results and discussion

Experimental setup

The hardware setup used for the proposed work consists of Intel Pentium 5 processor 11th Generation based machine with Graphics Memory of 4GB. The storage capacity is around 1 Tera Byte (Solid State). The deep learning experiments were run with Keras V2.15 library python, with TensorFlow application. Jupyter Platform V7.2 was used for the experimentation of the deep learning models. The experimentation used TUP virtual



Fig. 7. Learning Rate Curve for the Exponential Decay model for optimization.

Epoch	Step-Loss	Validation-Loss	Accuracy	Validation-Accuracy
1	0.0161	0.0054	0.9940	0.9978
2	0.0040	0.0029	0.9989	0.9990
3	0.0008	0.0030	0.9998	0.9998
4	0.0008	0.0030	0.9998	0.9998

Table 4. Inception V3 results.

processors of TensorFlow library in online platform. The system graphical memory of 2GB was additionally used for the enhancing the image processing capabilities.

The models used were InceptionV3, VGG-16, ResNet-50, VGG-19, CNN-MLP and InceptionV2ResNet. The learning rates were triggered properly using the optimizer at every epoch during the callback process. The learning rate was almost constant for every other model except InceptionV2ResNet, where the learning rate reduced as every epoch increase, with the influence of the exponential decay function as represented in the Fig. 7. The performances of the models are recorded for every epoch for a batch size of 438 samples for all the models, 875 for CNN-ResMLP and 1000 for InceptionV2ResNet.

Dataset description

The dataset contains images of various samples of the concrete surfaces with and without crack. The image data are divided into two as negative (without crack) and positive (with crack) in separate folder for image classification. Each class has 20000images with a total of 40,000 images with 227×227 pixels with RGB channels. The dataset is generated from 458 high-resolution images (4032×3024 pixel) with the method proposed by[53]. High resolution images found out to have high variance in terms of surface finish and illumination condition. No data augmentation in terms of random rotation or flipping or tilting is applied. The dataset is acquired from the source[54].

Results

This section contains the results of the various imagenet models that were used for the comparative analysis of building cracks. The description follows,

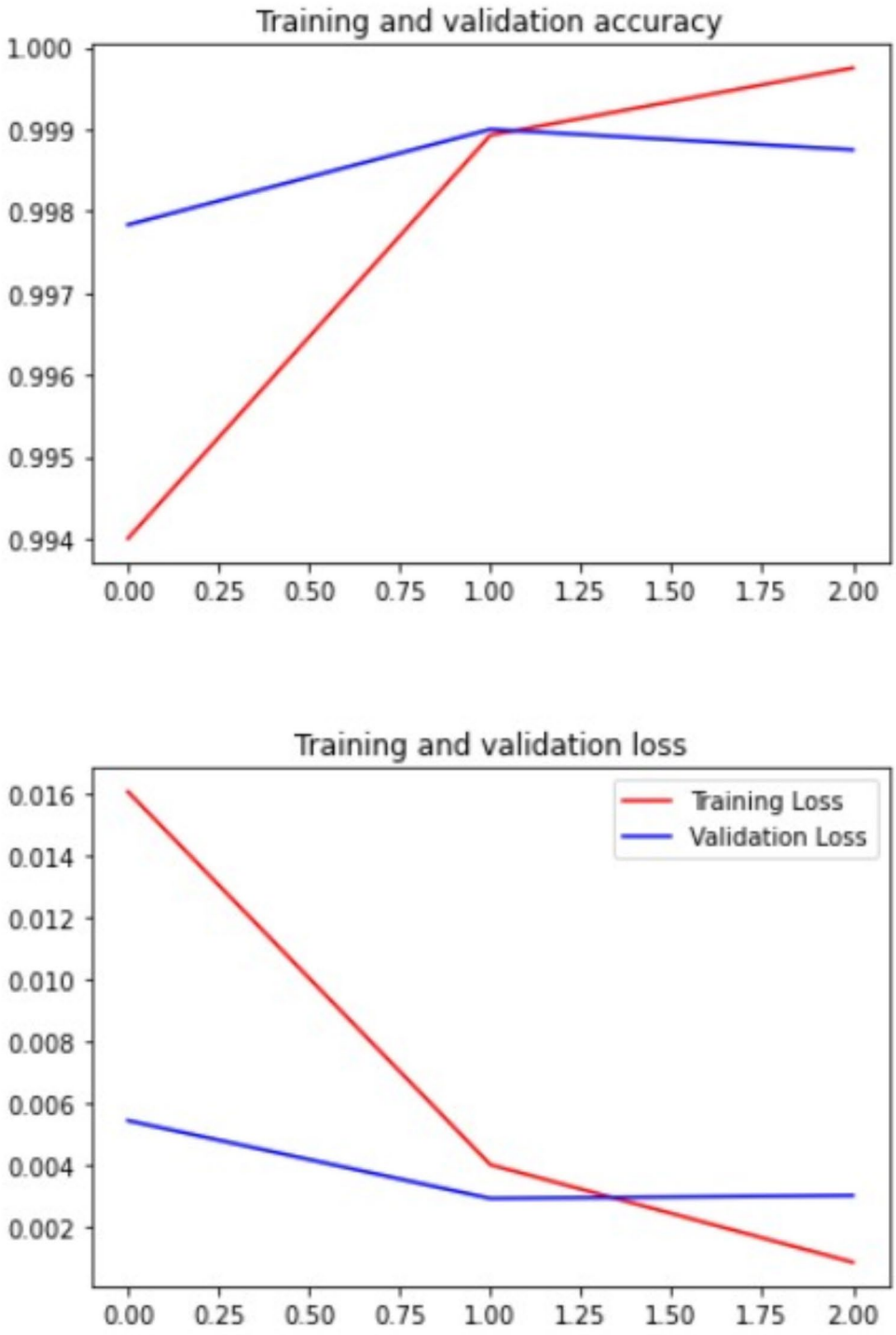


Fig. 8. Experimental Analysis of Inception V3 for Accuracy and Loss Estimation.

Inception V3

The model has been trained and converged in four epochs where it attained the maximum accuracy of 0.9998. The loss function has been gradually improving from 0.0161 to 0.0008 because of the Adam optimization and Max-Polling. Thus, Inception V3 provides best solutions of all the methods for crack detection. The results are presented in Table 4. The model response is measured as per the Fig. 8.

Epoch	Step-Loss	Validation-Loss	Accuracy	Validation-Accuracy
1	0.0543	0.0241	0.9839	0.9932
2	0.0194	0.0206	0.9948	0.9947
3	0.0163	0.0210	0.9954	0.9946
4	0.0130	0.0961	0.9966	0.9795
5	0.0113	0.0160	0.9969	0.9964
6	0.0099	0.0318	0.9974	0.9937

Table 5. VGG-16 results.

VGG-16
The model has been trained and converged in four epochs where it attained the maximum accuracy of 0.9974. The loss function has been gradually improving from 0.0543 to 0.0099 because of the RMSProp and Avg-Polling. The VGG-16 results are presented in Table 5. The model response is measured as per the Fig. 9.

ResNet-50
The model has been trained and converged in four epochs where it attained the maximum accuracy of 0.9998. This is almost equal to Inception V3, but the validation accuracy 0.9989. The loss function has been gradually improving from 0.0256 to 0.048 because of the SGD and Avg-Polling. The ResNet-50 results are presented in Table 6. The model response is measured as per the Fig. 10.

CNN + ResMLP
CNN + ResMLP provides linear connectivity with fully connected layer. There is no pooling, batch normalization, transform layers and it is a simple feed forward network. It provides cross-channel entropy and supported by GELU (Gaussian Error Linear Unit) for activation. This method provides accuracy of 0.9915 and sparse categorical accuracy of 0.9919. This model provides the step loss of 0.264 and validation loss of 0.268.

The results are tabulated in Table 7 and the response of CNN + ResMLP is plotted in Fig. 11.

VGG19
The model has been trained and converged in four epochs where it attained the maximum accuracy of 0.9969. The loss function has been gradually improving from 0.0684 to 0.0110, because of the RMSProp and Avg-Polling. The VGG-19 results are presented in Table 8. The model response is measured with Training and Validation Accuracy, and Training and Validation loss as per the Figs. 12 and 13.

InceptionV2ResNet
InceptionV2ResNet can process input image feature dimensions of 299*299*3, but in the proposed work this algorithm effectively processed images in the feature map dimensions of 224*224*3 as well. The performance of the model is depicted in the Fig. 14. The model produced the Accuracy, Precision, Recall and F1-score of 99.7%, with AUC of 99.99%. These metrics are superior, when compared with the existing metric evaluation of all other models.

Discussion
The confusion matrix was built on the classification models to check the number of false positives and false negatives. This experiment is conducted based on the classification accuracy, true positives, true negatives, false positives and false negatives measurement. There are 12,000 samples of crack images which are classified, either as crack or non-crack images. The confusion matrix for the Inception V3 model is presented in Fig. 15. There are no false positives for the no-crack and 2 false positives in the crack detection, the classification accuracy of the model is 0.9998. The confusion matrix of the VGG-16 model is presented in the Fig. 16. This model has 23 false positives for no crack and 8 false negatives for the crack images, with the accuracy of 0.9974. The confusion matrix of ResNet-50 is presented in Fig. 17.

The next representation is the confusion matrix of the model ResNet-50. The accuracy of the model is similar to Inception V3, which is 0.9998. The classification produced one false positive and one false negative in each class as represented in Fig. 17. The next model in which the confusion matrix applied is CNN-MLP. This is presented in Fig. 18. This model produced 68 mis-classified instances for class 0 and 25 mis-classified instances for class 1 with the accuracy of 0.9917. The confusion matrix of the VGG-19 model is presented in Fig. 19. This model achieved accuracy of 0.997 with 9 outliers for class 0 and 27 outliers for class 1, where class 0 is no-crack images and class 1 represent crack images. The confusion matrix of the Inception V2 ResNet is presented in Fig. 20. The model achieved accuracy 0.997 with 36 false positives and True negatives, where 11 instances are true negatives and 25 false positives. All the models were executed successfully with all the performance metrics are evaluated and recorded.

Comparative analysis
The Inception V3 model provides best of accuracy and ROC with 0.9998 and loss of 0.0008. Hence, it is the best of all other models that are used for the crack detection. The next best model is ResNet-50 with loss value of 0.0048 and accuracy of 0.9998. The validation accuracy of this model is only 09988. Hence this model comes next to Inception V3. VGG-16 comes third with accuracy of 0.9974 with loss value of 0.0099. Finally, CNN + ResMLP

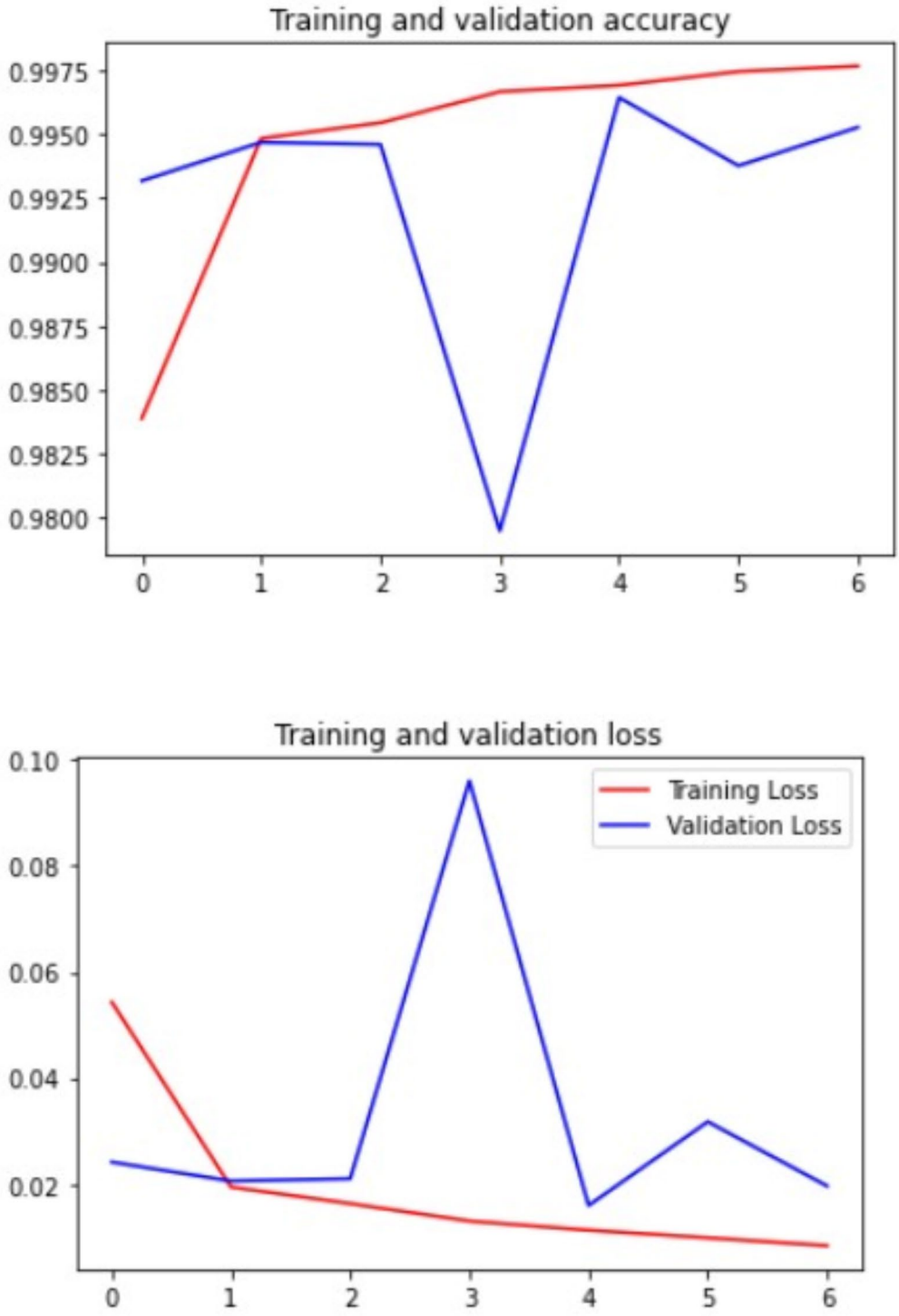


Fig. 9. Experimental Analysis of VGG-16 with Accuracy and Loss Estimation.

with accuracy of 0.9919 and loss of 0.0268 are also tabulated. The comparative analysis of various deep learning models used in the proposed work is presented in Fig. 21 and the results are tabulated in Table 9.

The metrics used in this proposed work are accuracy, validation accuracy, loss and validation loss. In general, the metrics are specified similarly for the training and validation. Hence forth the accuracy function of the deep image neural models can be expressed as per the Eq. 8.

$$Accuracy = TP + TN/(TP + TN + FP + FN) \tag{8}$$

Epoch	Step-Loss	Validation-Loss	Accuracy	Validation-Accuracy
1	0.0256	0.0101	0.9923	0.9973
2	0.0084	0.0070	0.9980	0.9983
3	0.0068	0.0061	0.9985	0.9985
4	0.0060	0.0054	0.9986	0.9987
5	0.0055	0.0049	0.9987	0.9987
6	0.0051	0.0046	0.9999	0.9989
7	0.0048	0.0044	0.9998	0.9989

Table 6. ResNet-50 results.

Where T_P is true positive T_N is true negative and F_P is false positive F_N is false negative. The loss value of the deep image neural network can be expressed as per the Eq. 9.

The comparative analysis based on the performance in detection of the crack for the various ImageNet models are presented in the Table 9. The Inception V3 and ResNet-50 model have performed better in the measurements of the Precision and Recall respectively. Both the models however achieved equal amount of the accuracy.

The InceptionV2ResNet provided the best of the ROC, which is important for the validation data, where produced value of 0.9999. These models were recorded with best performances in the respective criteria of the evaluation. The performance comparative analysis is presented in the Fig. 22.

$AverageLoss = -(y \log(p) + (1-y) \log(1-p))$ (9).

The w^T is the weight function $Y(i)$ is the actual predicted loss and $y^{[\cdot}(i)]$ is the deviated loss from the original value.

Challenges

- **Sensitivity**: Getting enough sensitivity to spot tiny cracks in reinforced concrete structures is one of the difficulties. Standard methods frequently aren't sensitive enough to spot cracks in their early stages, which can eventually grow larger and cause more severe structural damage. The usage of Ultra-High-definition cameras may capture precise crack images, with the available sampling and pre-processing techniques that may support the detection of sensitive cracks. Table 10 presents the comparative analysis of performance metrics used for crack detection.

- **Integration** The integration of the smart materials with the reinforced concrete structure is another issue. The complexity of reinforced concrete buildings and the requirement to ensure that the smart elements are effectively bound to the concrete make this challenging. Surface testing is the solution for this challenge, and prior to this construction baseline planning is essential.
- **False Positives** With smart material crack detection methods, false positives might be an issue. This could be as a result of environmental factors like temperature changes or vibrations, which can generate signals resembling those generated by real fissures. The false positives can be minimized with improved learning rate and training accuracy. K-fold validation reduces over-fitting of models that also minimizes the false positive measurements.
- **Durability** It is crucial to make sure that the intelligent materials employed for crack detection are durable. Structures made of reinforced concrete, which may be subject to adverse environmental conditions, should pay particular attention to this. Environment protection coating is the solution in addressing this challenge. The object detection methods also can check the durability of the building surfaces.
- **Picture Quality**: There are issues in the calibration of the crack detection photos. When cracks are difficult to discern from other image features due to poor image quality, false positives or false negatives may result. Usage of UHD cameras with better preprocessing techniques support picture quality and the crack detection.
- **Picture Threshold** Choosing an adequate threshold value that can identify cracks from other features in the image might be a difficult process.
- **Visual examination** This is a subjective technique that has historically been used to identify cracks in concrete. Visual inspection frequently results in mistakes and inaccuracies. The visual inspection is extended and supported by the proposed crack detection techniques.
- **Computational Complexity** Processing huge data sets can demand a significant number of computational resources, and image-based fracture detection algorithms can be computationally challenging. The computational complexity of the models are reduced by using the k-fold approximation, residual blocks, and reduction of the grid size.
- **Detection of thin propagating cracks**: Finding tiny spreading cracks can be difficult since they can be hard to spot in concrete photographs because they can look similar to other features. Incorporation of the object detection along with image classification techniques can support the detection of the thin propagation cracks.

Solutions

The proposed work is not a replacement, but surely as a better solution with models like InceptionV2ResNet with AUC of 0.9999 assures, reliability, durability and with 1000 classifiers and 210 MB 3 layer deep convolution model it handles computational complexity and can also be used in handheld devices unlike UNET or YOLO8 models. This model has accuracy of 0.997 which is really good for detection of the thin cracks and performs the

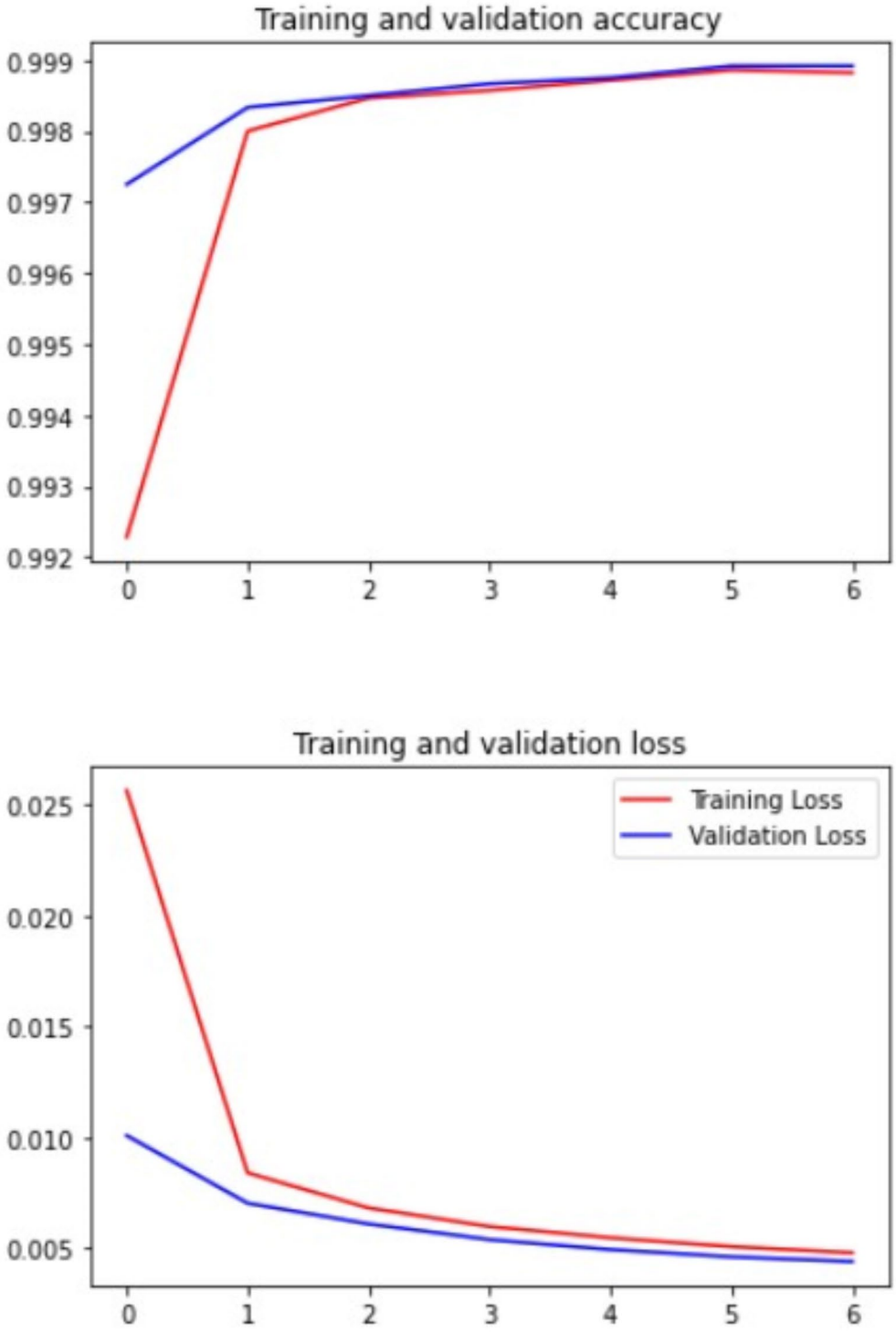


Fig. 10. Experimental Analysis of ResNet-50 with Accuracy and Loss Estimations.

visual examination with the higher accuracy levels. The models are optimized with appropriate optimizers to achieve maximum accuracy; hence they possess improved sensitivity and reduced false-positives.

- **Data Augmentation**: With the increase of the number of labelled images, data augmentation techniques like flipping, rotating, and adding noise to images can enhance a CNN's performance.
- **TransferLearning**: Pre-trained CNNs can be modified using transfer learning to perform concrete fracture detecting tasks. Using pre-existing models and datasets can result in time and resource savings.

Epoch	Step-Loss	Validation-Loss	Sparse-cat-Accuracy	Val-Accuracy
1	0.0570	6.0964	0.9814	0.9915
2	0.0319	0.0452	0.9906	0.9898
3	0.0268	0.0264	0.9919	0.9915

Table 7. CNN + ResMLP results.

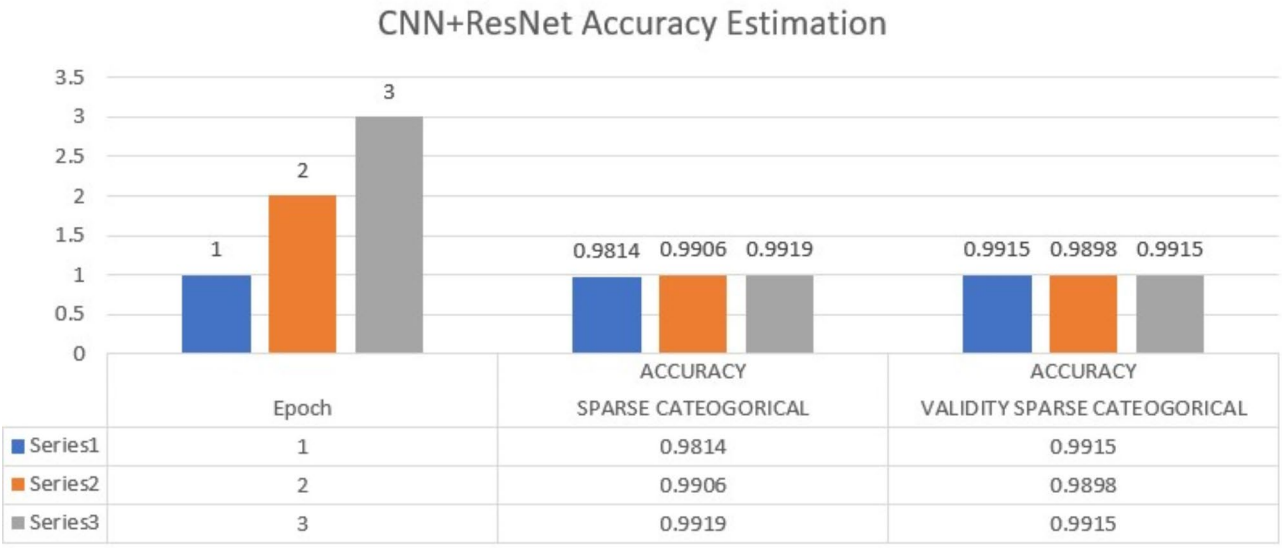


Fig. 11. Experimental Analysis of CNN + ResMLP with Accuracy and Loss Estimations.

Epoch	Step-Loss	Validation-Loss	Accuracy	Validation-Accuracy
1	0.0682	0.0243	0.9807	0.9927
2	0.0202	0.0197	0.9941	0.9940
3	0.0160	0.0192	0.9955	0.9948
4	0.0167	0.0178	0.9952	0.9952
5	0.0133	0.0357	0.9965	0.9921
6	0.0129	0.0274	0.9963	0.9931
7	0.0110	0.0209	0.9970	0.9948

Table 8. VGG19 results.

- **Hyperparameter adjustment**: For modification of variables like the learning rate, batch size, and number of layers, hyperparameter tuning can be utilized to improve the performance of a CNN.
- **Ensemble Learning**: To increase the precision of fracture detection, multiple CNN models can be combined using ensemble learning approaches.
- **Post-processing methods**: Morphological processes, for example, can be utilized to enhance the precision of thin propagating crack detection results and to fine-tune crack identification results.
- **Standardization**: The creation of standardized crack detection techniques can aid researchers in comparing and evaluating various methodologies and enhance the reproducibility of findings.
- **Computational Efficiency**: Improving the computational efficiency of image based crack detection methods can speed up processing and save time and resources.
- **Generalization**: This experimental setup and the crack detection framework is generally can be applied to the similar computer vision problems, provided the adequate quality and feature size of the images and the target classification is binary. This model must be revised for multi-class classification.

Potential research findings
These findings are observed as the potential research outcomes as follows,

- The proposed work shows improved stability and performance using hybrid ImageNet models such as InceptionV2ResNet. This provides AUC of 0.9999. This model was not used in the existing systems.

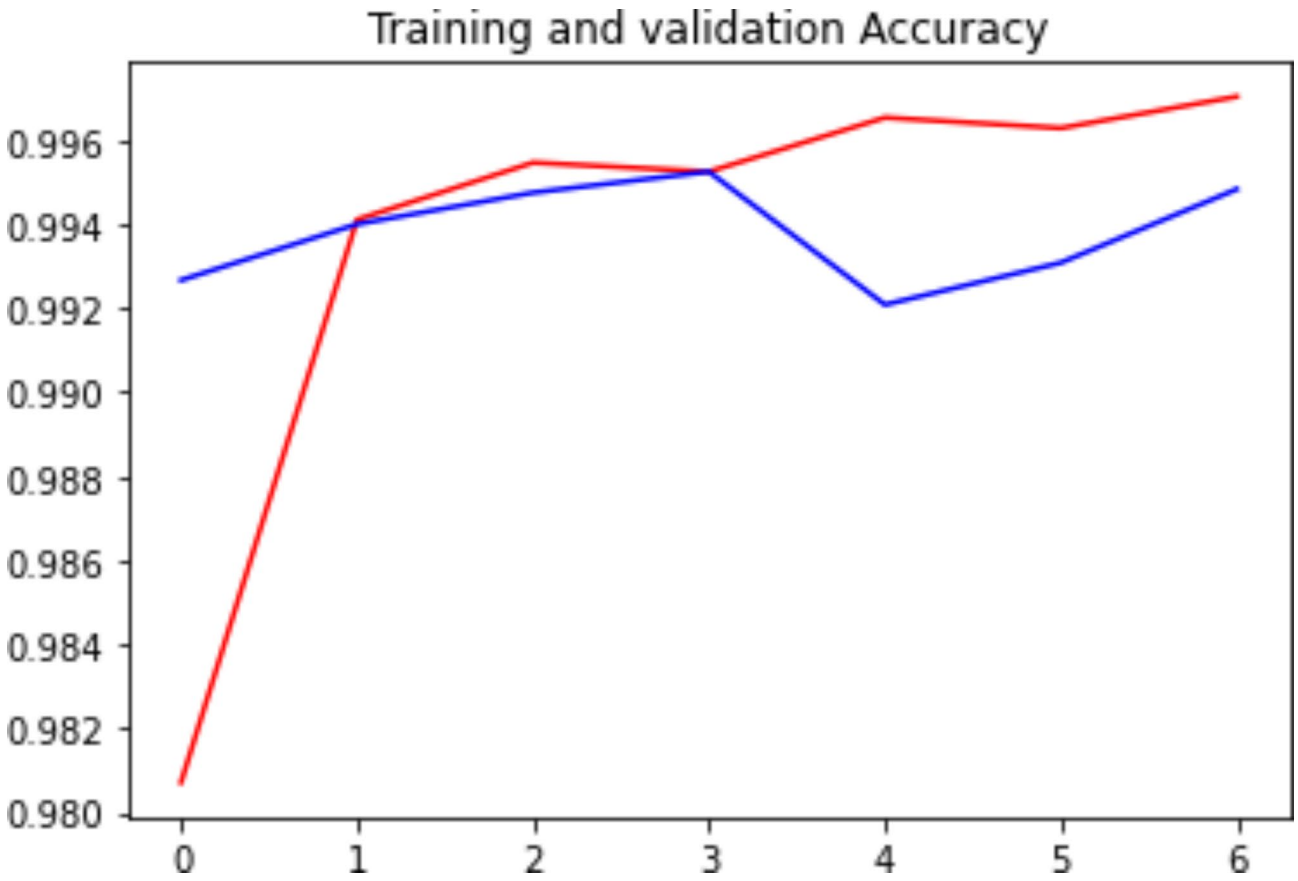


Fig. 12. Experimental Analysis of VGG-19.

- The proposed work triggered the learning rate and callback from the optimizers, which shows significant improvement on the model evolution.
- The selection of the appropriate optimizers increases the performance of the model for crack detection.
- The complete evaluation of the model parameters and performance metrics are necessary to arrive any solution on the comparative analysis, such as Training and Validation parameters (loss and accuracy) with performance metrics such as Accuracy, Precision, Recall and F1-Score.The proposed work analysed all these parameters and metrics.
- The proposed models perform better than the previous works in terms of classification metrics, for example with Accuracy values of InceptionV3 and ResNet-50 such as 0.9998 with better Recall and Precision such as 0.9999 respectively.

Excavation study

- Excavation study with Artificial intelligence is mainly done in three parameters such as, Ground Input parameters, wall input parameters and support input parameters[55].
- The data relevant to the excavation process is mainly measured from the groundwater drawdown, wall and ground displacement, and finally due to wall and support member force.
- Three generic modules such as ground water module, displacement module and wall module interfaces are developed with deep learning models.
- Parameters like shear force, wall displacement, bending moment and strut axial force are measured from the wall side modules.
- These parameters derive the stability of the wall structure during the excavation process with the help of Artificial neural network if the data is numeric and with the deep image neural network model if the data items are images.

Geotechnical evaluations of the crack detection

- There are many reasons that could cause cracks in buildings. There are geographical reasons such as swelling soil, shrinkage in the building materials, inappropriate building structures with poor load-bearing capacity.
- The evaluation process is mainly based on the quality of the building materials, building drawing evaluation, structural and foundational capacity of the building, and the soil quality estimates.
- The quality of the concrete and the steel bar is evaluated by the rebound hammer test and tensile strength test.

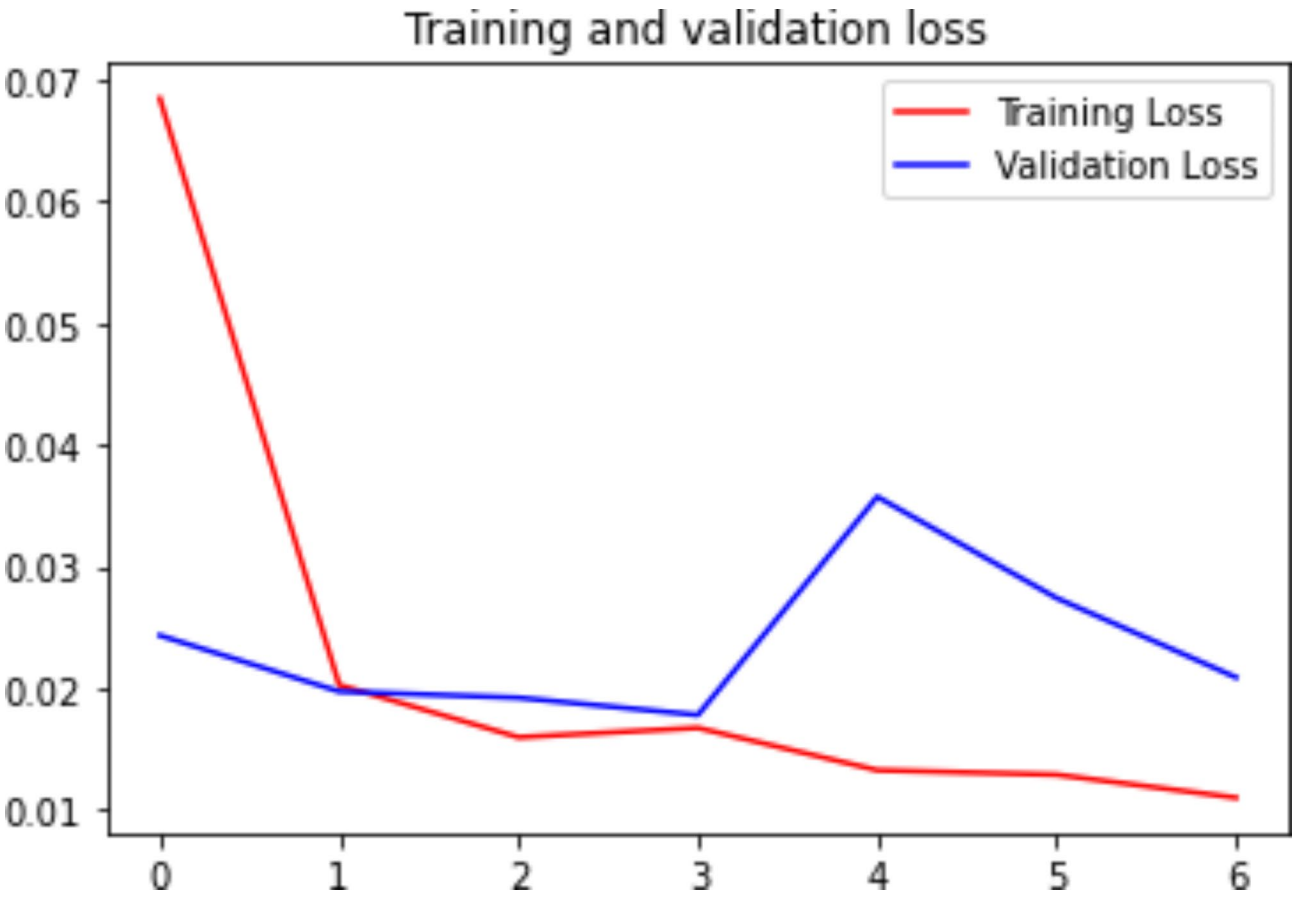


Fig. 13. Experimental Analysis of VGG-19.

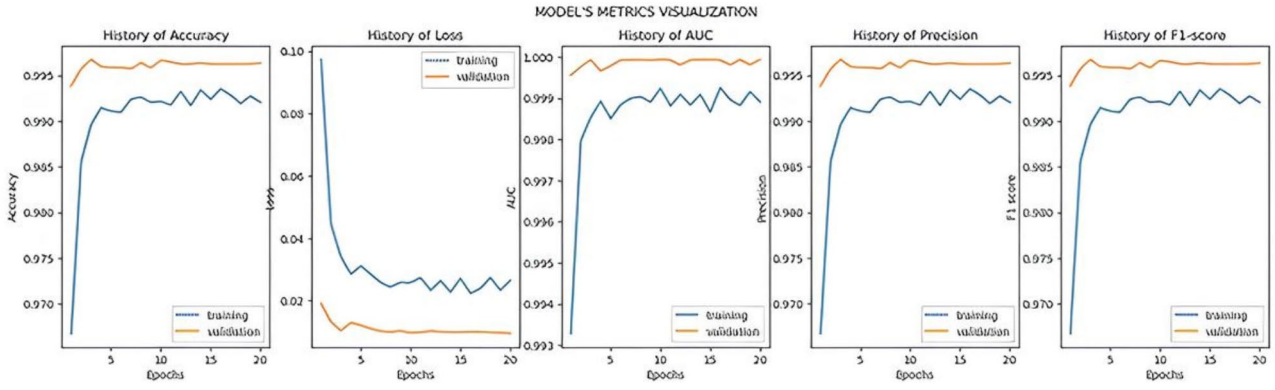


Fig. 14. Experimental Analysis of InceptionV2ResNet with all the Hyperparameters.

- The building drawings are used for the evaluation whether the existing structure would be able to support the new building.
- Cone penetration Test (CPT), boring log and soil penetration test along with the built drawings were used for the evaluation of the foundation capacity and quality of the soil beneath the building.
- The wall stress analysis helps to identify the cause of the crack using the finite element program.

Discussion on the deep learning models

- The InceptionV3 model has best accuracy production probability, with lesser size, when compared to the ResNet 50, VGG16 and VGG-19. It has both the advantages of time and space.

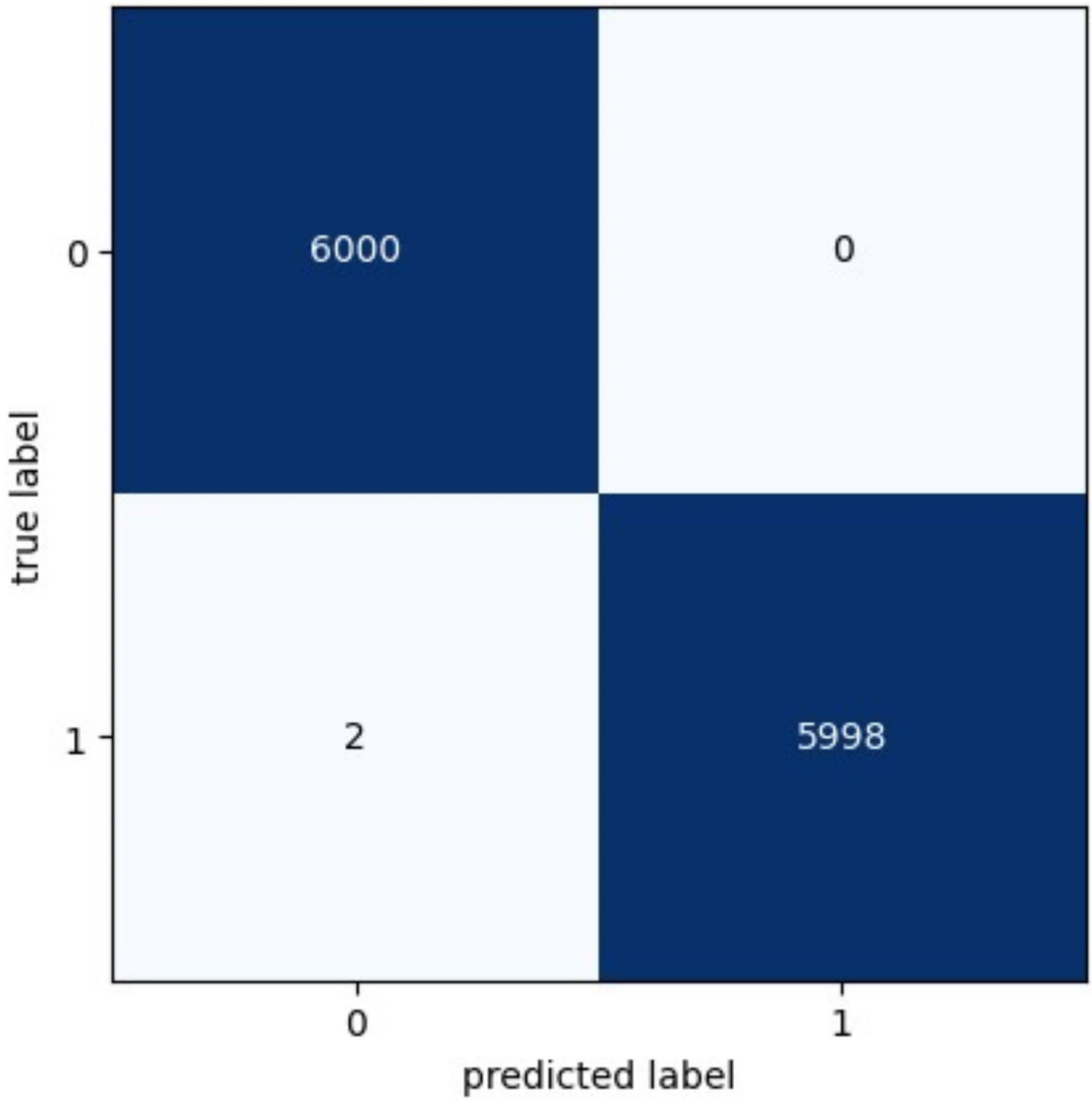


Fig. 15. Confusion Matrix for Inception V3 model.

- ResNET 50 uses the global average polling rather than the max pooling. Hence it has the ability to provide deeper convolutions when compared to VGG-16 and VGG-19 with the compact size.
- The benefits of these two models are combined in InceptionV2ResNET model which has the accuracy and lightweight capabilities of Inception and deeper convolution capabilities of ResNET.
- VGG-16 and VGG-19 are easily trainable, but heavier and takes more time to train. The capabilities of crack detection is checked on these models to understand the stability of the dataset and quality of the images acquired, and how these images are classified by various models. The availability of the models must not influence the quality of the crack detection. Thus a clear examination and comparative analysis is carried out in the proposed work.
- The performance metrics are marginal, but crack detection is a life saving application, it requires extreme evaluation of cracks, hence the best possible model can serve in any area of crack detection as per the analysis of the proposed work.
- The object-detection models like YOLO[56] can also be integrated into the deep learning models. In such cases very minor cracks can also be detected, if new images are presented for testing. This would also enhance the

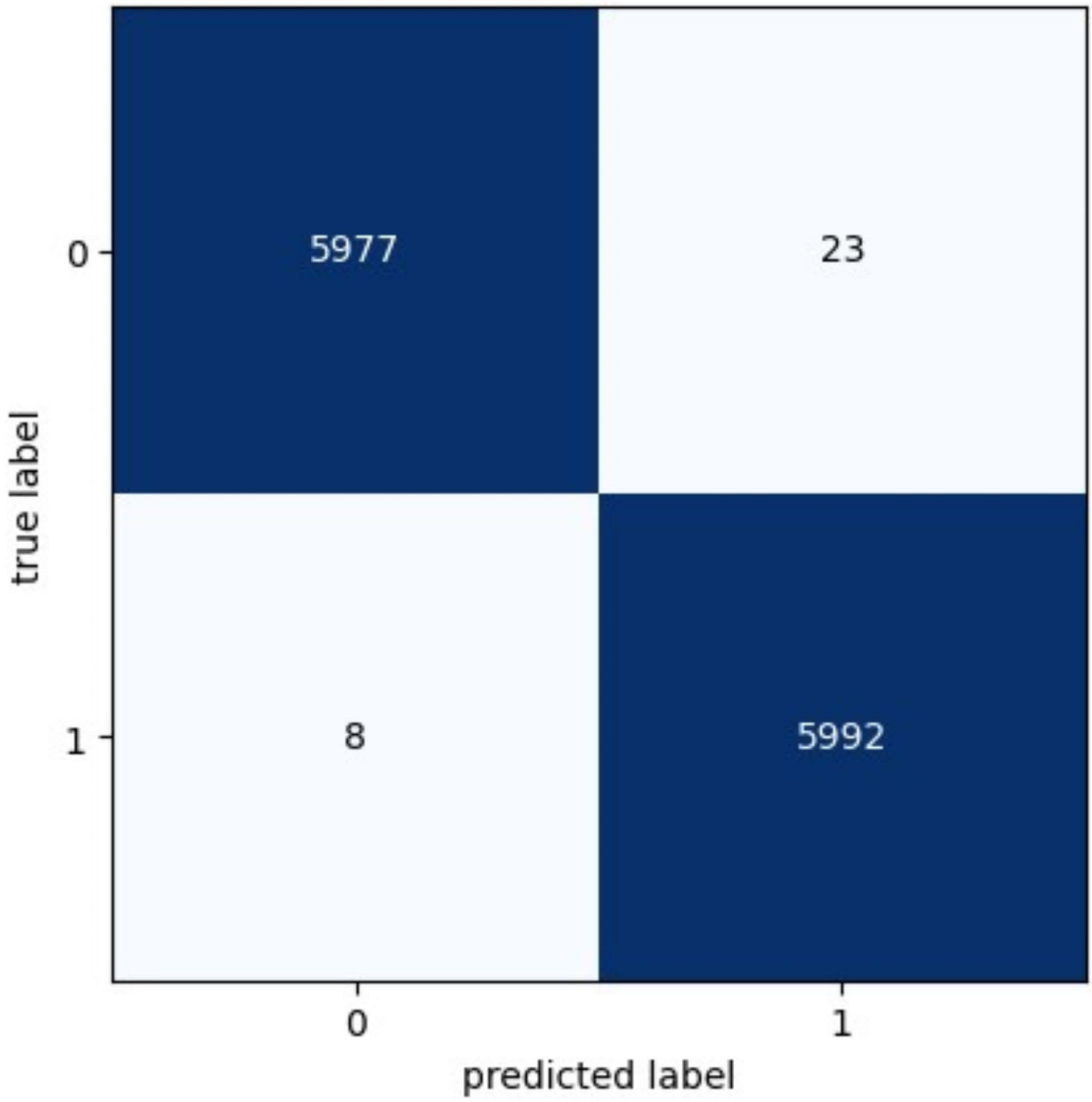


Fig. 16. Confusion Matrix for VGG-16 model.

reliability of the proposed system. This can be carried out in the further research, since the proposed work has different parameters for comparison.

Case studies

Cracks in public buildings and their remedies

Non-destructive testing techniques are used to determine the structure's strength in the first place so that it may be determined if it will be useful or need to be dismantled. Jiwaji University's political science, economic, and archaeology buildings were the sites of a few case studies that were done at various public buildings in Gwalior, MP. Moti Mahal structure. The Nagaji Institute of Technology and Management (NITM) building was reported to have cracks[57], and they were unsure if it would still function or if it should be dismantled solution to it was to hire structural engineer who will assess the results of non-destructive tests. Here, distinguishing non-structural surface cracks from those posing a threat to the building's stability was a significant challenge Advanced non-destructive testing technique, including as radiography and ultrasonic testing, were used to remedy issue. By evaluating the intrinsic integrity of the building materials, these systems were able to precisely distinguish between surface cracks that were cosmetic and those that could have structural consequences. This strategy

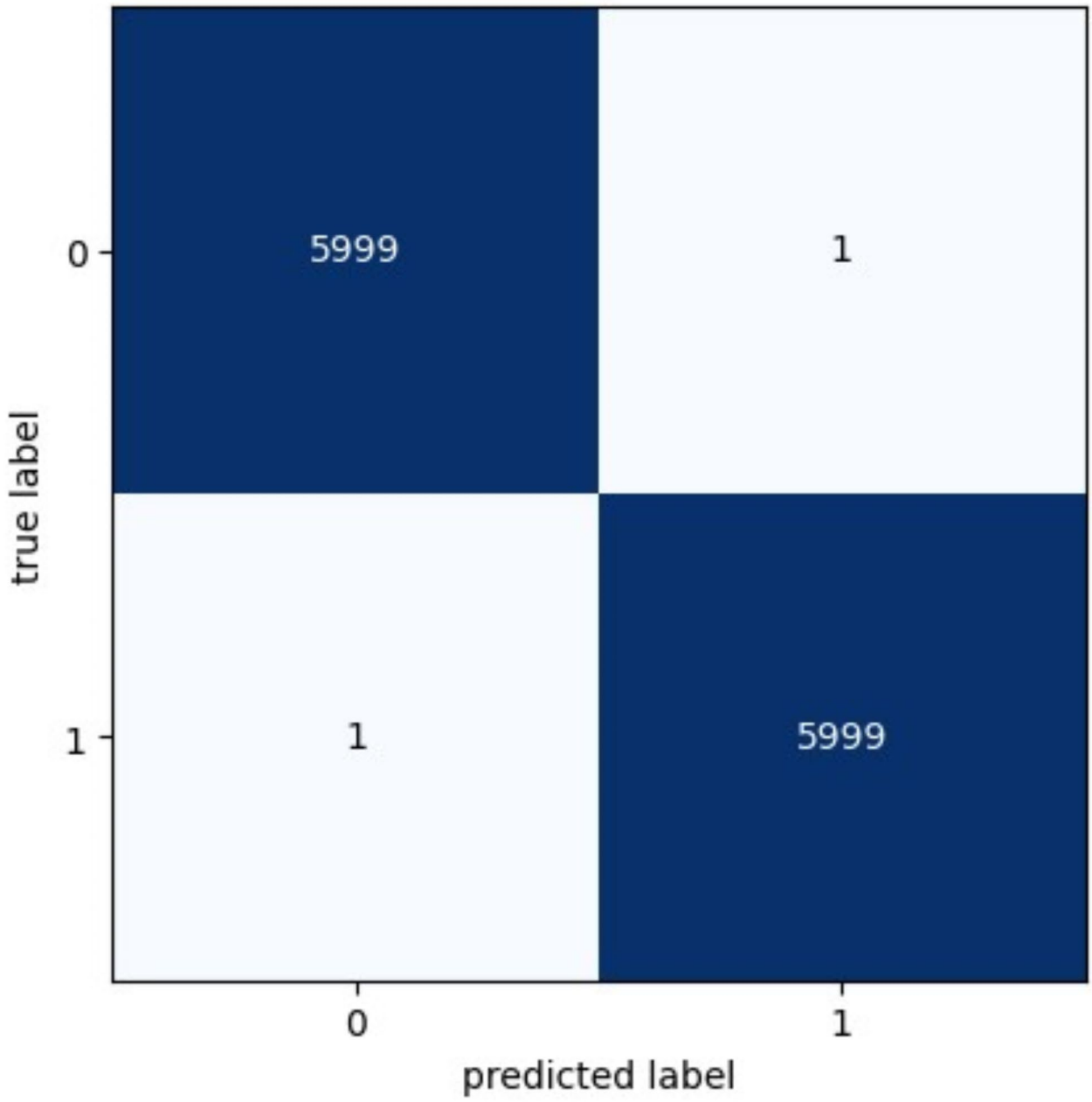


Fig. 17. Confusion Matrix for ResNet-50 model.

made sure that the nature and severities of the fractures were precisely understood, directing the right actions for the best possible structural maintenance.

A case study of Dr.Egbogha building, University of Ibadan, Nigeria

Upon the discovery of extensive wall cracking during the Egbogha building investigation[58], a comprehensive geotechnical examination was conducted to investigate the characteristics of the soil. Based on the data, it was determined that the soil had a low specific gravity of 2.48 to 2.63, a liquid limit of 26–53%, an ideal moisture content of 13.50–26.20%, and was settling at a depth of 1.5 m. A high liquid limit, uneven settling, and potential foundation issues were some of the issues noted. They were tackled through the application of a multifaceted strategy. Stabilization techniques such as foundation reinforcement and soil improvement were implemented based on geotechnical engineering concepts. Soil conditioning and drainage systems were two moisture control techniques used to lessen the effects of the high liquid limit. Continuous observation was instituted to guarantee the efficacy of the executed remedies. By using a methodical approach, settlement issues were effectively reduced, halting more structural damage and guaranteeing the Egbogha building's long-term stability. To preserve structural integrity and quickly handle any unforeseen difficulties, regular monitoring is being conducted.

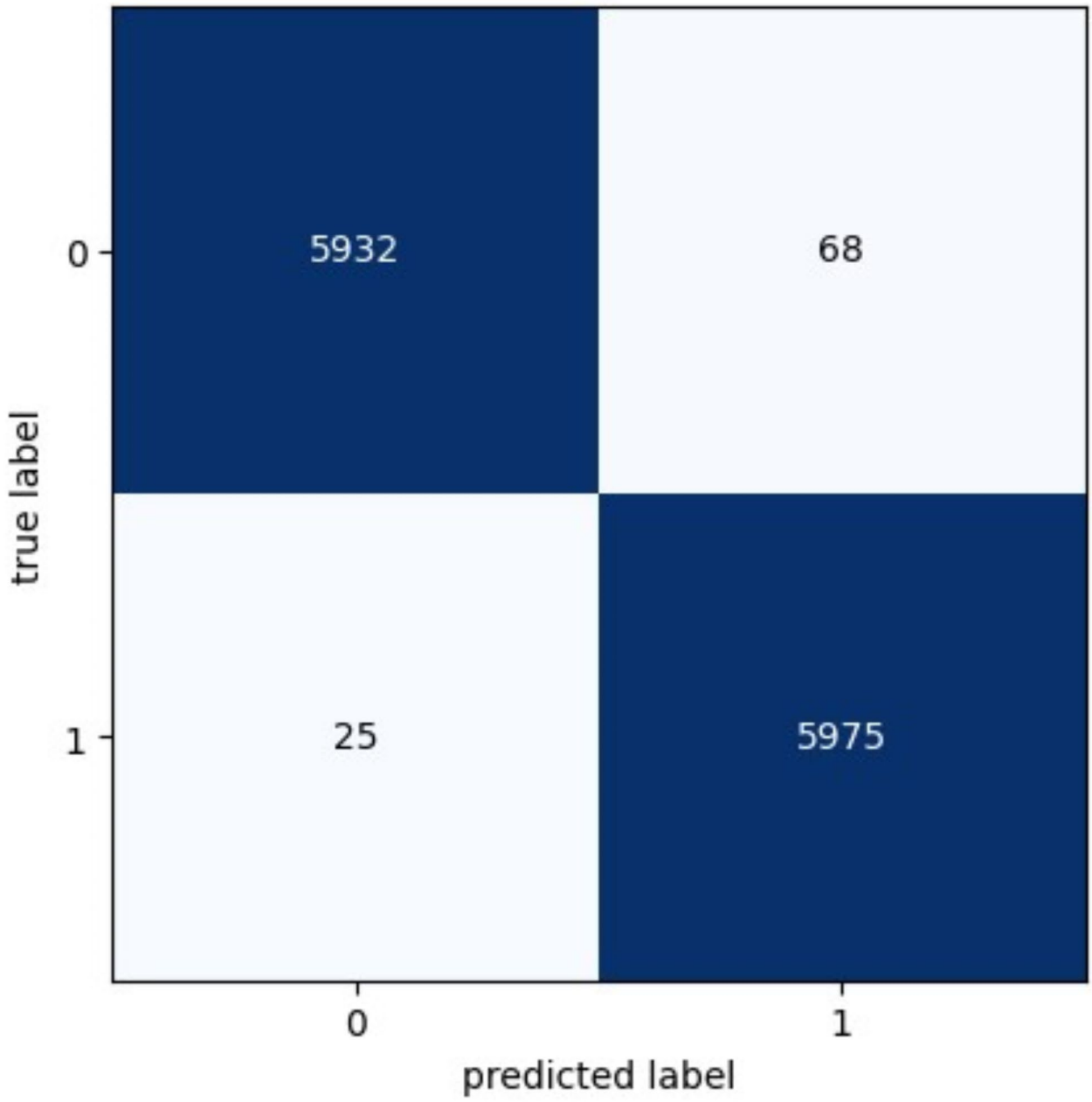


Fig. 18. Confusion Matrix for CNN-MLP model.

A case study in Vietnam

This study[59] looks into the behaviour of deep excavations in a Vietnamese sand layer and how they affect buildings in the vicinity. The study starts with the calibration of soil stiffness parameters for finite element analysis (FEA) based on observed diaphragm wall displacements. Additional challenges covered in the study include understanding ground surface deformation, validating criteria for assessing damage potential to nearby structures, and accurately assessing the impact of groundwater. Efficient groundwater management via methods like dewatering, careful FEA calibration, and ongoing observation of ground surface deformation and structural behaviour are all necessary for finding solutions. Data analysis, criterion validation, and the use of adaptive measures—such as modifications to excavation methods and structural reinforcement—are all part of the resolution process. The result highlights a thorough comprehension of the impact of excavations, validated assessment criteria, and effective adaptive measures. It also highlights the importance of a methodical approach, real-time monitoring, and adaptability in the management of deep excavations in Vietnam's difficult geological conditions.

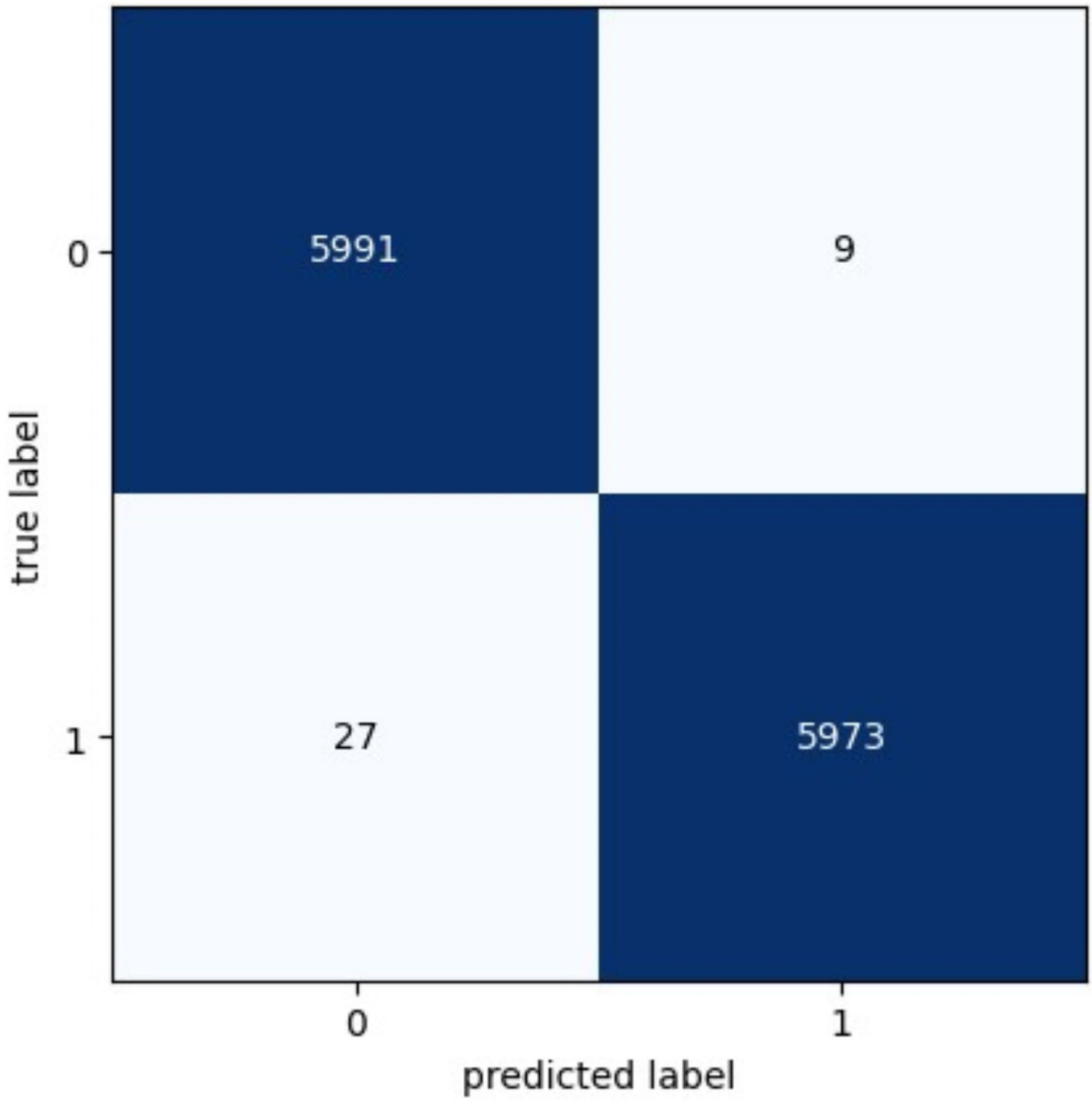


Fig. 19. Confusion Matrix for VGG-19 model.

Six crucial techniques for non-destructive testing crack testing

In recent years, non-destructive material testing has solidified its place in process monitoring. This inspection method's effective operation[1] has made it essential for thorough process control. At the same time, testing a component's geometric dimensions alone is no longer adequate in many businesses. This is explained by the fact that testing the components concurrently with continuous operation is becoming more and more important. Therefore, far more thorough testing is necessary to validate the appropriate qualities over the course of the service life.

Cracks visible on the material's surface are considered classical component faults. This is due to the fact that under constant strain, even the smallest cracks—which at first seem to be harmless—can expand and widen. In the worst case, a component failure may result from this. For example, the STATOGRAPH testing device from FOERSTER[37] is typically used to confirm this. The component and material, test frequency, and probe type all affect how big of a fault can still be fixed using a STATOGRAPH. Even fissures with a depth of only 30 µm can be discovered with the ideal design.

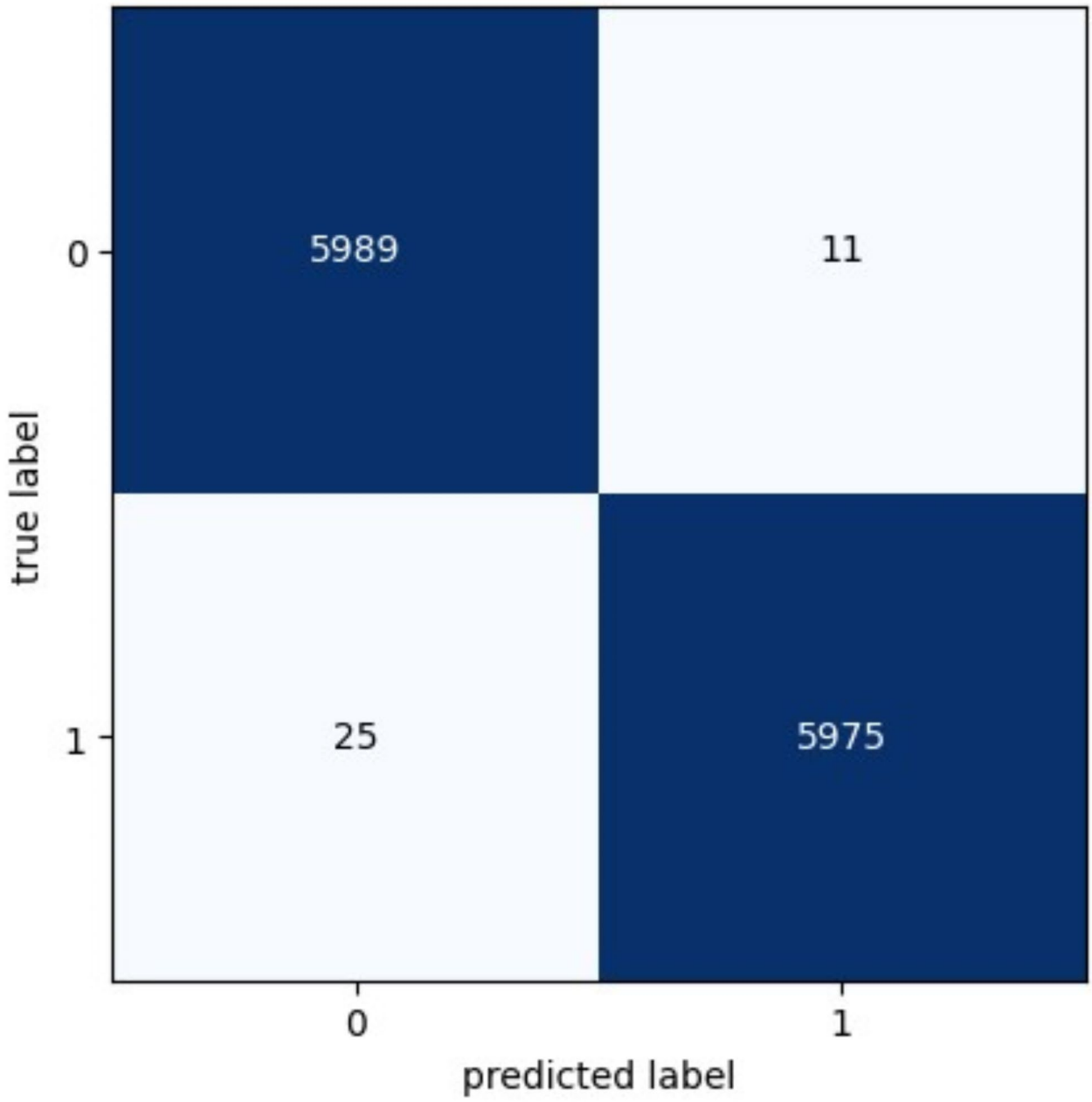


Fig. 20. Confusion Matrix for InceptionV2ResNet model.

Conclusion and future scope

The proposed work analyses various deep learning image techniques for crack detection to avoid further catastrophic damages to existing cracks which were detected. This is only a secondary support in detection of the crack, where the human intervention is not constantly available. The real-time manual inspection and repairing is always irreplaceable. The deep learning image techniques used in the proposed work were VGG16, CNN + ResMLP, InceptionV3, Inceptionv2ResNet, VGG-19 and ResNet-50. The InceptionV3 model rendered the best accuracy of 99.98% along with ResNet50. But the loss function of InceptionV3 is 0.0008 when compared with ResNet50 which is 0.048. Similarly, validation accuracy is 99.98 for InceptionV3 and 99.89 for ResNet50. Thus, we can conclude that InceptionV3 outperforms all other deep learning image models for crack detection. InceptionV2ResNet produced ROC of 0.9999 which is the best ROC value produced indicates the stability of the model to handle false positives. The Precision of the InceptionV3 is the best in the class and Recall of the ResNet50 is the best of all the models recorded such as 0.9998 and 0.9997 respectively. The convolution of InceptionV2 model and the kernel structure enhanced the balance of the classification.

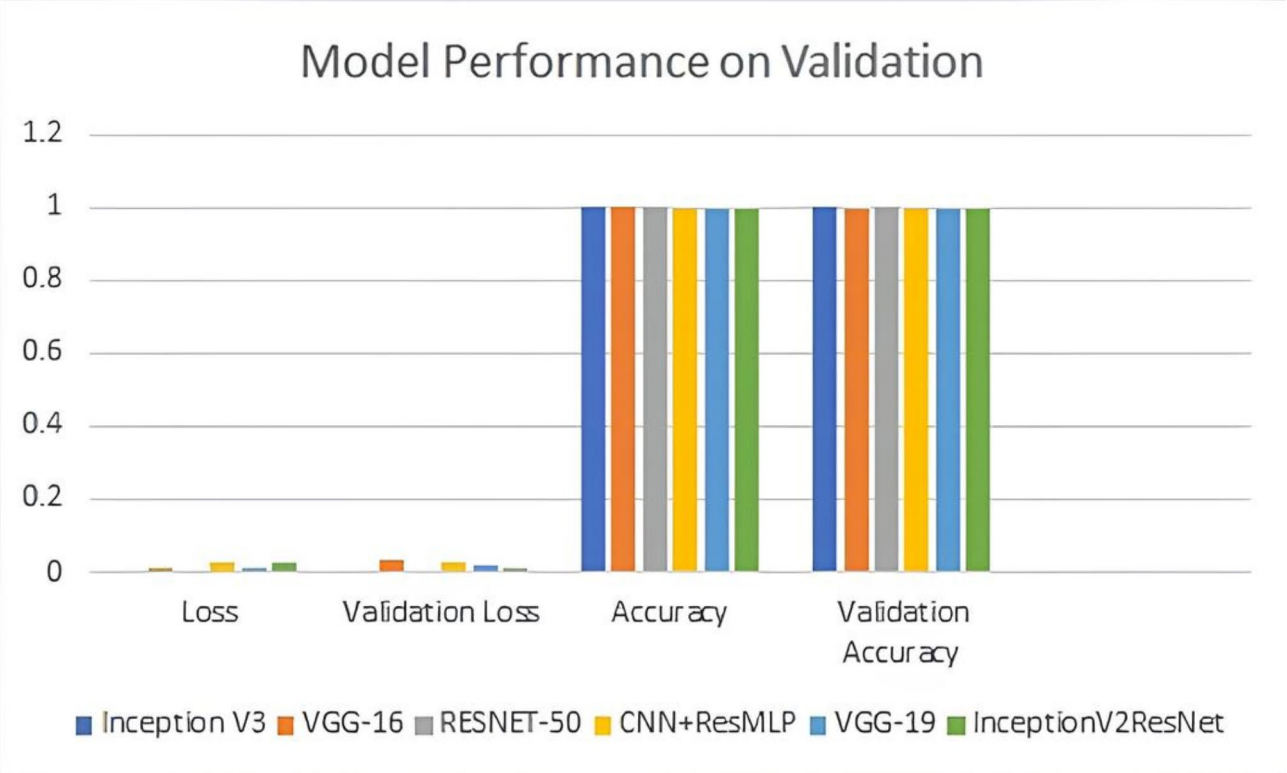


Fig. 21. Comparative Analysis of Various Deep Imagenet models for Crack Detection based on Validation.

Model	Loss	Validation-Loss	Accuracy	Validation-Accuracy
Inception V3	0.0008	0.0030	0.9998	0.9998
VGG-16	0.0099	0.0318	0.9974	0.9937
Resnet	0.0048	0.0044	0.9998	0.9989
CNN + ResMLP	0.0268	0.0264	0.9923	0.9915
Inception V2 ResNet	0.0265	0.0095	0.997	0.9964
VGG19	0.0110	0.0209	0.997	0.9948

Table 9. Comparative analysis of various deep Imagenet models for Crack Detection.

- Thus it is better to apply InceptionV3 for accuracy dependent applications, where the remedial solutions are required based on the efficiency of the crack detection.
- The InceptionV2ResNet may be applied in situations where image stability has the influence on the crack identification and prediction.

The Federated learning model frameworks can increase the parallel processing allows the possibility of data-fusion from various deep learning models or the ensemble deep learning models for crack detection with enhanced performance and accuracy.

The proposed model has limitations in explaining what features affect the classification performance of the model. The features that are related to the loss functions have not been identified and addressed. In order to improve the explainability related to the performance of the models, Explainable AI model can be used in future works to analyse the various parameters affecting the crack detection process with the illustration of the classification process. Federated learning can be deployed to optimize the performance when converging different AI models. The Explainable image learning models(XAI) will examine the features acquired from the images and would present the significance of every feature contributes to the output. It would support the decision-making process, along with the deep learning results obtained from the proposed models. This would be the future of crack detection with image processing.

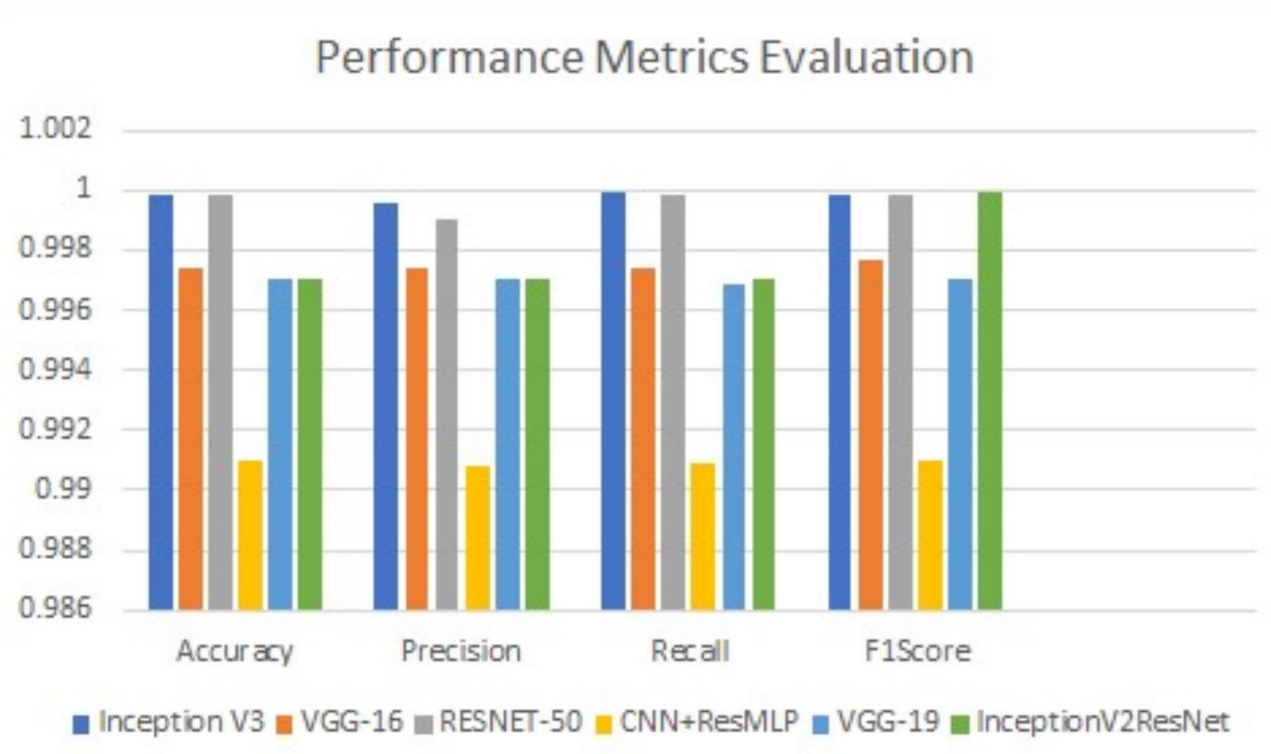


Fig. 22. Comparative Analysis of the Performance Metrics used for crack detection.

Model	Accuracy	Precision	Recall	F1-Score
Inception V3	0.9998	0.9999	0.9996	0.9998
VGG-16	0.9974	0.9961	0.9987	0.9974
ResNet-50	0.9998	0.9998	0.9998	0.9998
CNN + ResMLP	0.9923	0.09887	0.9958	0.9921
Inception V2 ResNet	0.997	0.998	0.997	0.9997
VGG19	0.997	0.996	0.9981	0.997

Table 10. Comparative analysis of the Performance Metrics used for crack detection.

Data availability

Authors declare that all the data being used in the design and production cum layout of the manuscript is declared in the manuscript.

Received: 1 September 2024; Accepted: 7 January 2025
Published online: 16 January 2025

References

1. Hiller, V. The 6 most important methods for crack testing in non-destructive material testing. Blog.foerstergroup.com. (2020)., August 20 https://blog.foerstergroup.com/en/component-testing/the-6-most-important-methods-for-crack-testing-in-non-destructive-material-testing
2. Satapathy, A., Dhale, Y., Patnaik, S. & Meena, T. Case study on cracks and its paraphernalia. Materials Today: Proceedings 45, 3560–3563 (2021). https://doi.org/10.1016/j.matpr.2020.12.1078. International Conference on Advances in Materials Research – 2019.
3. Prakash, U. S. & Nemmadi Comprehensive Guide on Cracks in Buildings: Causes, Prevention, and Repair. (2024)., August 22 https://nemmadi.in/cracks-in-buildings-causes-prevention-and-repair/
4. Causes and Remedies of Cracks in Concrete Buildings. The Constructor. (2014)., June 12 https://theconstructor.org/concrete/building-cracks-causes-remedies/5392/
5. https://www.ibm.com/topics/deep-learning
6. Golding, V. P., Gharineiat, Z., Munawar, H. S. & Ullah, F. Crack detection in concrete structures using deep learning. *Sustainability* **14** (13), 8117 (2022).
7. Laxman, K. C., Tabassum, N., Ai, L., Cole, C. & Ziehl, P. Automated crack detection and crack depth prediction for reinforced concrete structures using deep learning. *Constr. Build. Mater.* **370**, 130709 (2023).

8. Mayya, A. et al. Efficient hybrid ensembles of CNNs and transfer learning models for bridge deck image-based crack detection. In *Structures* **64**, 106538 (Elsevier, 2024).
9. Al-Huda, Z. et al. A hybrid deep learning pavement crack semantic segmentation. *Eng. Appl. Artif. Intell.* **122**, 106142 (2023).
10. Ghannadiasl, A. & Ghaemifard, S. Meta-heuristic algorithms: an appropriate approach in crack detection. *Innovative Infrastructure Solutions*. **9** (7), 263 (2024).
11. Urkude, N., Hora, M. S. & Utkarsh Enhancing machine learning models with metaheuristic Optimization techniques for accurate prediction of PSC in FRP-reinforced concrete slabs. *Mech. Adv. Mater. Struct.* 1–17. https://doi.org/10.1080/15376494.2024.2394985 (2024).
12. Pan, Y. & Zhang, L. Roles of artificial intelligence in construction engineering and management: a critical review and future trends. *Autom. Constr.* **122**, 103517 (2021).
13. Parhi, D. R. & Choudhury, S. Smart crack detection of a cracked cantilever beam using fuzzy logic technology with hybrid membership functions. *J. Eng. Technol. Res.* **3** (8), 270–278 (2011).
14. Silva, W. R. L. D. & Lucena, D. S. D.:Concrete cracks detection based on deep learning image classification. In *Proceedings* (Vol. 2, No. 8, p. 489). MDPI, (2018), June.
15. Hoang, N. D. Detection of surface crack in building structures using image processing technique with an improved Otsu method for image thresholding. *Adv. Civil Eng.* **2018** (1), 3924120 (2018).
16. Kapadia, H., Patel, P., Patel, J. & Kanani, N. Implementation of computer vision technique for crack monitoring in concrete structure. *J. Institution Eng. (India): Ser. A*. **104** https://doi.org/10.1007/s40030-022-00695-5 (2022).
17. Mohan, A. & Poobal, S. Crack detection using image processing: a critical review and analysis. *Alexandria Eng. J.* **57** https://doi.org/10.1016/j.aej.2017.01.020 (2017).
18. Dung, C. & Le Duc, A. Autonomous concrete crack detection using deep fully convolutional neural network. *Autom. Constr.* **99**, 52–58. https://doi.org/10.1016/j.autcon.2018.11.028 (2018).
19. Nishikawa, T., Yoshida, J., Sugiyama, T. & Fujino, Y. Concrete crack detection by multiple sequential image filtering. *Comp. -Aided Civil Infrastruct. Eng.* **27**, 29–47. https://doi.org/10.1111/j.1467-8667.2011.00716.x (2012).
20. Cheon, M. H., Hong, D. G. & Lee, D. *Surface crack detection in concrete structures using image processing* (Springer, Berlin, 2019).
21. Cho, S., Park, S., Cha, G. & Oh, T. Development of image processing for crack detection on concrete structures through terrestrial laser scanning associated with the octree structure. *Appl. Sci.* **8**, 2373. https://doi.org/10.3390/app8122373 (2018).
22. Yamaguchi, T. & Hashimoto, S. Fast crack detection method for large-size concrete surface images using percolation-based image processing. *Mach. Vis. Appl.* **21**, 797–809. https://doi.org/10.1007/s00138-009-0189-8 (2010).
23. Wang, P. & Huang, H. Comparison analysis on present image-based crack detection methods in concrete structures. In: 2010 3rd International Congress on Image and Signal Processing, vol. 5, pp. 2530–2533 (2010). https://doi.org/10.1109/CISP.2010.5647496.
24. Zhao, W., Liu, Y., Zhang, J., Shao, Y. & Shu, J. Automatic pixel-level crack detection and evaluation of concrete structures using deep learning. *Struct. Con. Health Monit.* **29**(8) e2981 (2022).
25. Yamane, T. & Chun, P. J. Crack detection from a concrete surface image based on semantic segmentation using deep learning. *J. Adv. Concr. Technol.* **18**(9) 493–504 (2020).
26. Arbaoui, A., Ouahabi, A., Jacques, S. & Hamiane, M. Concrete cracks detection and monitoring using deep learning-based multiresolution analysis. *Electronics* **10**(15), 1772 (2021).
27. Ko, P., Prieto, S. & Soto, B. Abecis: an automated building exterior crack inspection system using uavs, open-source deep learning and photogrammetry. (2021). https://doi.org/10.22260/ISARC2021/0086
28. Qayyum, W., Ehtisham, R., Camp, C., Mir, J. & Ahmad, A. Detecting cracks with Convolution Neural Network (CNN) with Variable image dataset. In *2nd International Conference on Recent Advances in Civil Engineering and Disaster Management* (pp. 166–170). Peshawar Pakistan, (2022).
29. Kolappan Geetha, G., Yang, H. J. & Sim, S. H. Fast detection of missing thin propagating cracks during deep-learning-based concrete crack/non-crack classification. *Sensors* **23**, 1–22. https://doi.org/10.3390/s23031419 (2023).
30. Munawar, H. S., Hammad, A., Haddad, A., Soares, C. & Waller, S. Image-based crack detection methods: a review. *Infrastructures* **6**, 115. https://doi.org/10.3390/infrastructures6080115 (2021).
31. Ali, L. et al. Performance evaluation of deep cnn-based crack detection and localization techniques for concrete structures. *Sensors* **21**, 1688. https://doi.org/10.3390/s21051688 (2021).
32. Li, S. & Zhao, X. High-resolution concrete damage image synthesis using conditional generative adversarial network. *Autom. Constr.* **147** https://doi.org/10.1016/j.autcon.2022.104739 (2023).
33. Kim, A. R. Crack detection of concrete structure using deep learning and image processing method in geotechnical engineering. *J. Comput. Civil Eng.* **34** (2), 04019046 (2020).
34. Adam, E. & Ammayappan, S. Construction of accurate crack identification on concrete structure using hybrid deep learning approach. *J. Innovative Image Process.* **3**, 85–99. https://doi.org/10.36548/jiip.2021.2.002 (2021).
35. Nguyen Kim, C., Kawamura, K., Nakamura, H. & Tarighat, A. Automatic crack detection for concrete infrastructures using image processing and deep learning. *IOP Conf. Series: Mater. Sci. Eng.* **829**, 012027. https://doi.org/10.1088/1757-899X/829/1/012027 (2020).
36. Cha, Y. J., Choi, W. & Büyüköztürk, O. Deep learning-based crack damage detection using convolutional neural networks. *Computer-Aided Civil Infrastructure Eng.* **32** (5), 361–378 (2017).
37. Hacıefendioğlu, K. & Başağa, H. B. Concrete road crack detection using deep learning-based faster R-CNN method. *Iranian J. Sci. Tech. Trans. Civil Engg.* **46**(2) 1621–1633 (2022).
38. Yu, Y. Crack detection of concrete structures using deep convolutional neural networks optimized by enhanced chicken swarm algorithm. Advances in Civil Engineering (2021) (2021).
39. Ottoni, A. L. C., Souza, A. M. & Novo, M. S. Automated hyperparameter tuning for crack image classification with deep learning. *Soft. Comput.* **27** (23), 18383–18402 (2023).
40. Ottoni, A. L. C., Novo, M. S. & Oliveira, M. S. A Statistical Approach to Hyperparameter tuning of deep learning for construction machine classification. *Arab. J. Sci. Eng.* **49** (4), 5117–5128 (2024).
41. VDM. Types of Cracks in Buildings Causes and Solutions. VDM. (2024)., November 4 https://www.vdmconstructions.com/construction/types-of-cracks-in-buildings-causes-and-solutions/
42. R, A. P. Surface crack detection (2019). https://www.kaggle.com/datasets/arunrk7/surface-crack-detection
43. Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J. & Wojna, Z. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 2818–2826), (2016).
44. Karen, S. & Andrew Zisserman. &. Very Deep Convolutional Networks for Large-Scale ImageRecognition. *Arvix doi* (2015). :https://arxiv.org/abs/1409.1556
45. He, K., Zhang, X. & Jian Sun. Shaoqing Ren, &. Deep Residual Learning for Image Recognition.Arvix, doi: (2015). https://arxiv.org/abs/1512.03385
46. Keiron, O. S. & Ryan Nash. &. An Introduction to ConvolutionalNeuralNetworks.*Arvix,doi* (2015). :https://arxiv.org/abs/1511.08458
47. Behrad, F. & Saniee Abadeh, M. An overview of deep learning methods for multimodal medical data mining. *Expert Syst. Appl.* **200**, 117006. https://doi.org/10.1016/j.eswa.2022.117006 (2022).
48. Malitha, G., Norbert Zolek & Arvix Performance of Machine Learning Classification in Mammography Images using BI-RADS., *doi* (2023). :https://arxiv.org/abs/2311.08493

49. Szegedy, C., Ioffe, S. & Alex Alemi. Vincent Vanhoucke, &. Inception-v4, Inception-ResNet and the Impact of ResidualConnectionsonLearning.*Arvix,doi* (2016). :https://arxiv.org/abs/1602.07261
50. Victor Ikechukwu, A., Murali, S., Deepu, R. & Shivamurthy, R. C. ResNet-50 vs vgg-19 vs training from scratch: a comparative analysis of the segmentation and classification of pneumonia from chest x-ray images. *Global Transitions Proc.* **2** (2), 375–381. https://doi.org/10.1016/j.gltp.2021.08.027 (2021). International Conference on Computing System and its Applications (ICCSA2021).
51. Praveen, S. P., Srinivasu, P. N., Shafi, J., Wozniak, M. & Ijaz, M. F. Resnet-32 and Fastai for diagnoses of ductal carcinoma from 2d tissue slides. *Sci. Rep.* **12** (1), 20804. https://doi.org/10.1038/s41598-022-25089-2 (2022).
52. Silva, W. R. L. d. & Lucena, D. S. d. Concrete cracks detection based on deep learning image classification. *Proceedings* **2** (8). https://doi.org/10.3390/ICEM18-05387 (2018).
53. Zhang, L., Yang, F., Daniel Zhang, Y. & Zhu, Y. J. Road crack detection using deep convolutional neural network, *IEEE International Conference on Image* Processing *(ICIP)*, Phoenix, AZ, USA, 2016, pp. 3708–3712, (2016). https://doi.org/10.1109/ICIP.2016.7533052
54. Özgenel, Ç. Fırat Concrete Crack Images for Classification, *Mendeley Data, v2.* (2019).
55. Yoo, C., Aizaz, H. S., Abbas, Q. & Yang, J. Artificial Intelligence (AI)-based deep excavation designed Program. *J. Korean Geosynthetics Soc.* **17** (4), 277–292. https://doi.org/10.12814/JKGSS.2018.17.4.277 (2018).
56. Muhammad Hussain. YOLOv5, YOLOv8 and YOLOv10: The Go-To Detectors for Real-time Vision. Arvix,doi: (2024). https://arxiv.org/abs/2407.02988
57. Pathak, R. & Rastogi, D. Case study on cracks in public buildings and their remedies. *Int. J. Sci. Res.* **6**, 2319–7064 (2015).
58. Agbede, O., Jatau, N., Oluokun, G. & Akinniyi, B. Geotechnical investigation into the causes of cracks in building: a case study of dr.egbogha building, university of Ibadan, Nigeria. *Int. J. Eng. Sci. Invention*. **4** (2), 18–22 (2015).
59. Huynh, Q. T., Lai, V. Q., Boonyatee, T. & Keawsawasvong, S. Behavior of a deep excavation and damages on adjacent buildings: a case study in Vietnam. *Transp. Infrastructure Geotechnology*. **8** (3), 361–389 (2021).
60. He, Y. et al. Local–global feature adaptive Fusion Network for Building Crack Detection. *Sensors* **24** (21), 7076. https://doi.org/10.3390/s24217076 (2024).
61. Cao, J. et al. Automatic detection and assessment of crack development in ultra-high-performance concrete in the spatial and Fourier domains. *Front. Struct. Civ. Eng.* **18**, 350–364. https://doi.org/10.1007/s11709-024-1042-x (2024).
62. Qiqi Zeng, D. & Miao and Jianxin Ye. An Attention Crack Detection Network for Structural Health Monitoring. In Proceedings of the 2024 International Conference on Smart City and Information System (ICSCIS '24). Association for Computing Machinery, New York, NY, USA, 388–392. (2024). https://doi.org/10.1145/3685088.3685156

Author contributions
All authors contributed equally to this manuscript. All authors reviewed the manuscript.

Funding
The authors received no specific funding for this study.

Declarations

Conflict of interest
The authors declare that they have no conflicts of interest to report regarding the present study.

Additional information
Correspondence and requests for materials should be addressed to S.S.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2025